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2D Finite Element Analysis of a Microstrip Transmission Line

MTH321 Project 2

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Abstract

Microstrip impedance can be estimated from the capacitance of its cross-section, but a finite-element result also depends on the outer computational boundary and how the field near conductor corners is resolved. We studied a finite-thickness microstrip trace using a two-dimensional electrostatic model with conforming triangular P_1 elements.

After verifying the solver with a manufactured solution, we examined domain truncation and mesh convergence. The uniform-mesh energy error had a fitted rate of 0.623, below the smooth-problem rate of one. We initially tested coordinate-line grading near the corners, but an audit showed that the original cross-mesh error integration was not sufficiently reliable for precise efficiency claims. A separately tested triangle-overlay method was therefore used to compare uniform, graded and flux-jump adaptive refinement.

At a relative energy error of 0.30, adaptivity needed approximately 224 degrees of freedom (DOFs), compared with 273 for uniform refinement and 349 for grading. Its cumulative computation time was nevertheless higher.

The selected M7 solution gave a dielectric capacitance of 120.620 pF/m. Reusing exactly the same mesh for the all-air calculation gave 35.9682 pF/m, an effective permittivity of 3.3535 and a characteristic impedance of 50.642 Ω. Hammerstad–Jensen and IPC-2141 gave approximate impedance values of 50.201 Ω and 49.393 Ω, respectively.

The results distinguish improved mesh efficiency from reduced total runtime while retaining explicit numerical limitations.

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1 Introduction

A microstrip transmission line consists of a conducting signal trace above a conducting reference plane, separated by a dielectric substrate and surrounded by air. When the electrical length is not negligible compared with the relevant signal wavelengths, the characteristic impedance of the line controls reflection behaviour and therefore signal integrity. Under a lossless quasistatic or quasi-TEM model, this impedance can be inferred from the capacitance per unit length of the cross-section and the corresponding all-air capacitance[1]. The report structure follows the project requirement that numerical claims be supported by verification, convergence evidence, cost information, and reproducible artefacts.

We focused on two main questions: how reliable the calculated microstrip impedance is, and whether local mesh refinement can improve computational efficiency. To answer these questions, we first verified the FEM solver and studied domain and mesh convergence. We then compared uniform, graded and adaptive refinement using energy error, degrees of freedom, matrix nonzeros and measured computation time.

The nominal geometry is a copper trace of width $w = 3.0\,\mathrm{mm}$ and thickness $t = 0.035\,\mathrm{mm}$ on a substrate of height $H = 1.6\,\mathrm{mm}$ and relative permittivity $\varepsilon_r = 4.4$. All computations use SI units internally. The finite computational domain has $x \in [-W, W]$ and $y \in [0, H_{\mathrm{air}}]$; the final engineering domain uses $W/H = H_{\mathrm{air}}/H = 8$ after a separate domain study.

2 Physical and Mathematical Model

2.1 Geometry and material model

The substrate occupies $0 < y < H$ and has $\varepsilon_r = 4.4$. The air region is above the substrate and has $\varepsilon_r = 1$. The signal conductor is centred at $x = 0$ and occupies $H \le y \le H + t$ and $-w/2 \le x \le w/2$. Its interior is removed from the field domain, so its boundary is an internal Dirichlet boundary rather than a material subregion.

Table 1: Physical and computational parameters for the final microstrip model.

Quantity	Value	Meaning
H	$1.6 \times 10^{-3}\,\mathrm{m}$	substrate height
w	$3.0 \times 10^{-3}\,\mathrm{m}$	trace width
t	$3.5 \times 10^{-5}\,\mathrm{m}$	trace thickness
ε_r	4.4	substrate relative permittivity
V_0	$1\,\mathrm{V}$	trace potential
W/H	8	final half-width ratio
H_{air}/H	8	final height ratio

[1] For the lossless transmission-line and microstrip model, see D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley, 2011. The project-specific capacitance-pair identities are stated in the supplied *Capacitor2d* specification, Appendix A.

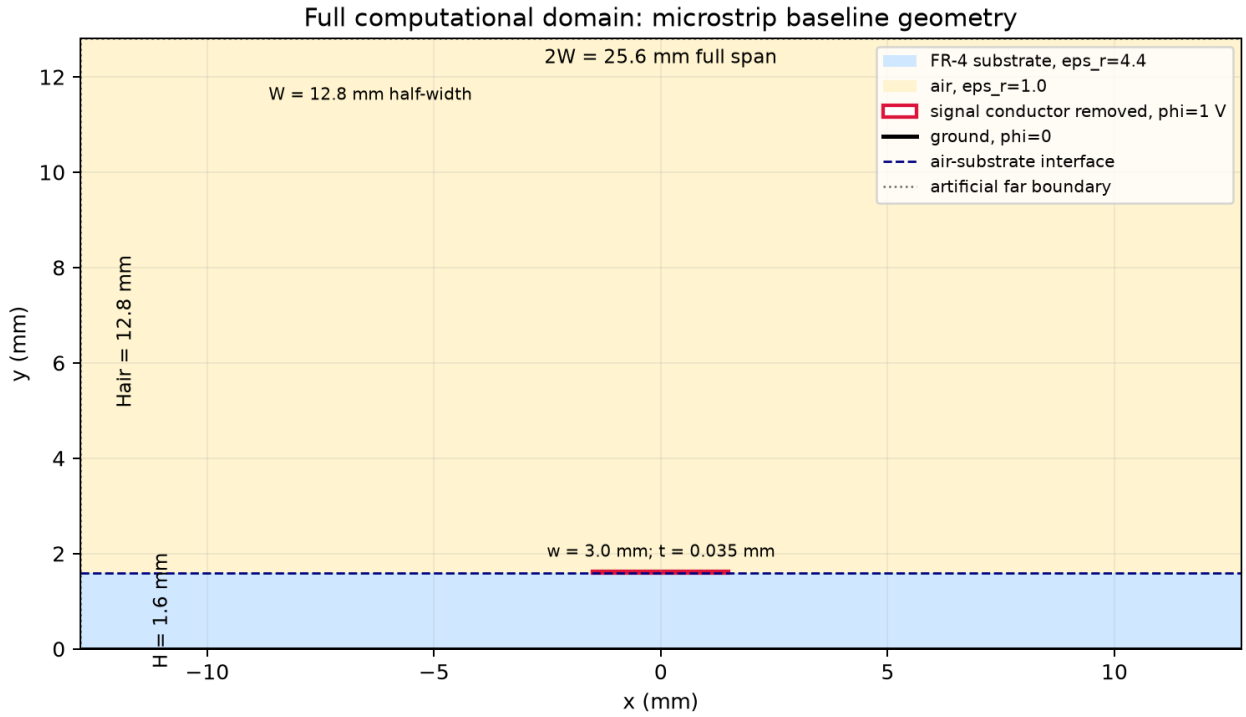


Figure 1: Final geometry convention. The domain half-width is W, so the full horizontal span is $2W$. The trace interior is removed from the field domain and the outer top and side boundaries are artificial truncation boundaries.

2.2 Governing equation and boundary conditions

The electrostatic energy model is described by Hiptmair [2, Sec. 1.2.2]. For an admissible potential, the stored energy per unit length is

$$W_e(\phi) = \frac{1}{2}\int_\Omega \varepsilon(x,y)|\nabla\phi|^2\, d\Omega.$$

With the conductor potentials fixed, its stationarity gives the following divergence-form equation. The permittivity is a material property, so it is piecewise constant here: $4.4\varepsilon_0$ in the substrate and ε_0 in air. Hackbusch [1, Sec. 10.1] explains why such positive discontinuous coefficients are naturally treated in a variational formulation, with separate integration over the material regions.

The potential ϕ satisfies

$$-\nabla\cdot(\varepsilon\nabla\phi) = 0 \qquad \text{in } \Omega, \tag{1}$$

where $\varepsilon = \varepsilon_0\varepsilon_r$ is piecewise constant. The essential boundary conditions are

$$\phi = V_0 \quad \text{on the signal trace,} \qquad \phi = 0 \quad \text{on the ground plane.}$$

The artificial top and side boundaries use the homogeneous Neumann condition

$$\varepsilon\nabla\phi\cdot n = 0.$$

This same far-boundary convention is used throughout the domain-truncation sequence. Across the air–substrate interface, the exact electrostatic interface conditions are continuity of potential and balance of normal displacement flux.

3 Finite Element Formulation

Let Γ_D denote the union of the trace and ground Dirichlet boundaries. The affine trial space is

$$V_g = \{u \in H^1(\Omega) : u = V_0 \text{ on the trace, } u = 0 \text{ on the ground}\},$$

and the homogeneous test space is

$$\mathcal{V} = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

Here V_0 denotes the prescribed trace voltage, while $\mathcal{V}$ denotes the homogeneous function space. Multiplying (1) by $v \in \mathcal{V}$ and integrating by parts gives

$$\int_\Omega \varepsilon \nabla\phi \cdot \nabla v \, d\Omega - \int_{\partial\Omega} \varepsilon \nabla\phi \cdot n \, v \, ds = 0.$$

The Dirichlet part of the boundary term vanishes because $v = 0$ there, and the artificial Neumann term is zero by construction. The weak problem is therefore:

$$\text{find } \phi \in V_g \text{ such that } \int_\Omega \varepsilon \nabla\phi \cdot \nabla v \, d\Omega = 0 \quad \forall v \in \mathcal{V}.$$

3.1 Well-posedness and discrete stability

The finite field domain is a connected bounded polygonal Lipschitz domain with the trace hole removed. Its ground and trace boundaries give Γ_D positive boundary measure. For

$$a(u, v) = \int_\Omega \varepsilon \nabla u \cdot \nabla v \, d\Omega, \qquad \varepsilon_0 \le \varepsilon(x) \le 4.4\varepsilon_0,$$

Cauchy–Schwarz and the positive lower coefficient bound give

$$|a(u, v)| \le 4.4\varepsilon_0 \|\nabla u\|_{L^2} \|\nabla v\|_{L^2}, \qquad a(v, v) \ge \varepsilon_0 \|\nabla v\|_{L^2}^2. \tag{2}$$

For $v \in \mathcal{V}$, the Poincaré inequality $\|v\|_{L^2} \le C_P \|\nabla v\|_{L^2}$ holds because the Dirichlet boundary is nonempty with positive measure. Thus the gradient seminorm is a norm on $\mathcal{V}$, equivalent to its H^1 norm, and a is continuous and coercive. The dielectric jump preserves these properties because both materials have strictly positive permittivity.

Choose an H^1 lifting g of the prescribed trace and ground values and write $\phi = g + u$. The homogeneous problem is $a(u, v) = -a(g, v)$ for all $v \in \mathcal{V}$. Its bounded right-hand side and coercivity give existence and uniqueness by the Lax–Milgram theorem. In particular, the difference w of two solutions satisfies $a(w, w) = 0$, hence $\nabla w = 0$; connectedness makes w constant and its zero Dirichlet trace forces $w = 0$. The homogeneous Neumann outer boundary therefore does not leave the constant nullspace of a pure-Neumann problem.

The same estimate applies to the conforming subspace $\mathcal{V}_h \subset \mathcal{V}$. For a nonzero vector z of free coefficients with associated function v_h,

$$z^T K_{FF} z = a(v_h, v_h) > 0.$$

Consequently, the reduced matrix is symmetric positive definite. For a load perturbation $\delta\ell$ with unchanged Dirichlet data, testing the difference equation by δu_h gives

$$\|\nabla \delta u_h\|_{L^2} \le \varepsilon_0^{-1} \|\delta\ell\|_{\mathcal{V}'}, \qquad \|\delta\ell\|_{\mathcal{V}'} = \sup_{v \in \mathcal{V}\setminus\{0\}} \frac{|\delta\ell(v)|}{\|\nabla v\|_{L^2}}.$$

This is stability in a function-space norm on the fixed domain, not a mesh-independent Euclidean matrix condition number. It also does not imply a first-order energy convergence rate: approximation rates depend on solution regularity and mesh resolution. The measured $p_E \simeq 0.623$ is compatible with a stable problem whose corner/interface field is less regular than the smooth verification solution.

The well-posedness argument follows the standard variational theory for elliptic problems discussed by Hackbusch [1, Sec. 7.2] and Quarteroni [3, Secs. 3.4 and 4.2]. In our model, the permittivity is discontinuous across the air–substrate interface, but it remains strictly positive in both regions. Therefore, the discontinuity does not prevent the weak problem from being coercive. Quarteroni's Galerkin discussion also explains the positive-definite stiffness matrix obtained after restricting to the homogeneous discrete test space.

3.2 Assembly, solution and capacitance extraction

The discrete space uses conforming triangular P_1 elements, as developed by Quarteroni [3, Sec. 4.5] and Hiptmair [2, Sec. 2.4]. Local element assembly and essential-boundary elimination follow the standard procedures described by Hiptmair [2, Secs. 2.7.4 and 2.7.6]. Writing

$$\phi_h = \sum_j \Phi_j N_j,$$

the stiffness matrix has entries

$$K_{ij} = \int_\Omega \varepsilon \nabla N_i \cdot \nabla N_j \, d\Omega.$$

Because ε is discontinuous at the material interface, the integral is evaluated element by element using each triangle's material tag. In a conforming H^1 finite-element space, ϕ_h is continuous across the air–substrate interface. Since the discontinuous coefficient is integrated in the same global weak form, the interface flux-balance condition is enforced automatically in the weak or distributional sense. No separate Lagrange multiplier is required. This does not mean that the two elementwise P_1 normal-flux traces $\varepsilon \nabla \phi_h \cdot n$ are pointwise identical on every interface edge.

The recorded environment uses scikit-fem 12.0.2 for mesh topology and the manufactured-solution basis, quadrature and assembly. In the microstrip solver, project code evaluates the constant P_1 basis gradients explicitly and assembles $K_{ij}^K = \varepsilon_K |K| \nabla N_i \cdot \nabla N_j$ into a SciPy sparse matrix. This expression integrates the element stiffness exactly because the coefficient and gradients are constant on each material-conforming triangle. SciPy supplies the sparse direct solve. Project code controls the trace hole, material and boundary tags, refinement, capacitance extraction and validation; these choices are checked separately from the library interfaces.

Dirichlet conditions are imposed by eliminating the trace and ground degrees of freedom, leaving a sparse symmetric positive definite free-degree system solved by a sparse direct solver.

The capacitance is extracted in two ways. The energy method is

$$W_e = \frac{1}{2} \int_\Omega \varepsilon |\nabla \phi_h|^2 \, d\Omega, \qquad C_E = \frac{2W_e}{V_0^2}.$$

The reaction-charge method sums the assembled Dirichlet reactions on the trace:

$$C_Q = \frac{|Q_{\text{trace}}|}{|V_0|}.$$

Agreement between C_E and C_Q checks consistency of energy integration and reaction extraction. Geometry and boundary-tag tests are also required, since a shared modelling error could affect both quantities. Spatial accuracy is assessed through refinement.

Algorithm 1 Microstrip P_1 FEM solver

Require: Material-conforming mesh $\mathcal{T}_h$, geometry, ε_K and voltage V_0

1: Generate the study mesh, or load the exact frozen mesh for a matched replay.
2: Verify the trace hole, interface, element regions and trace/ground/far boundary sets.
3: Set $D = D_{\text{trace}} \cup D_{\text{ground}}$, $F = \{\text{all nodes}\} \setminus D$.
4: Initialise $\Phi_D = V_0$ on the trace and $\Phi_D = 0$ on ground; set $K = 0$.
5: **for** each triangle $K_e \in \mathcal{T}_h$ **do**
6: Compute $|K_e|$ and its three constant basis gradients.
7: Assemble $\varepsilon_{K_e} |K_e| \nabla N_i \cdot \nabla N_j$ into the sparse matrix.
8: **end for**
9: Solve $K_{FF} \Phi_F = -K_{FD} \Phi_D$ by the sparse direct solver.
10: Evaluate $\nabla \phi_h$, W_e, $C_E = 2W_e/V_0^2$ and reactions $r = K\Phi$.
11: Sum trace reactions to obtain Q_{trace} and $C_Q = |Q_{\text{trace}}|/|V_0|$.
12: Check bounds, symmetry, residual, charge balance and C_E/C_Q consistency.

Ensure: Potential, capacitances and diagnostics; failed checks disqualify interpretation.

Algorithm 1 separates the physical boundary specification from the linear solve. The homogeneous Neumann edges add no load, whereas prescribed voltages enter through $-K_{FD}\Phi_D$. Reactions must be recovered from the full assembled matrix, before its boundary rows are discarded. The implementation is in `code/fem_solver.py` and `code/capacitance.py`.

4 Solver Verification and Baseline Checks

4.1 Manufactured-solution verification

Known-solution tests make the discretisation error measurable and allow observed rates to be compared with the smooth-solution estimates. This is the code-validation strategy discussed by Hiptmair [2, Sec. 3.8], with the expected finite-element energy and L^2 estimates given by Quarteroni [3, Sec. 4.5]. Matching the rates supports the tested implementation; it does not by itself verify every feature of the microstrip model.

Before using the microstrip result, the implementation was verified on a smooth known solution. The test problem uses a unit square with zero Dirichlet data and exact solution

$$u(x, y) = \sin(\pi x)\sin(\pi y).$$

The manufactured forcing and boundary data are explicitly

$$-\Delta u = f, \qquad f(x, y) = 2\pi^2 \sin(\pi x)\sin(\pi y), \qquad u|_{\partial(0,1)^2} = 0.$$

Here the named errors are $\|u-u_h\|_{L^2(\Omega)}$ and $\|u-u_h\|_E = (\int_\Omega |\nabla(u-u_h)|^2\, d\Omega)^{1/2}$, with coefficient one for this verification problem. The expected smooth P_1 rates are two and one, respectively. The five consecutive meshes below contain the required $h, h/2, h/4, h/8$ protocol and one further level $h/16$. Pairwise orders are $p_i = \log(E_{i-1}/E_i)/\log(h_{i-1}/h_i)$.

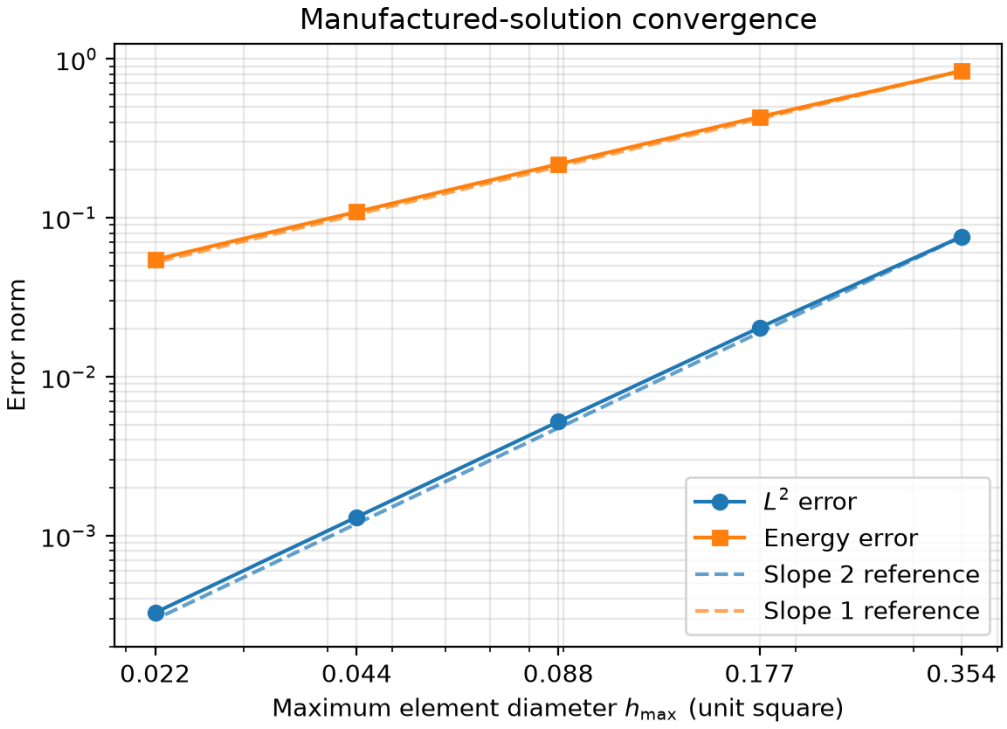


Figure 2: Manufactured-solution convergence. The finest-step observed orders are approximately 2.00 in L^2 and 1.00 in the energy norm, as expected for smooth P_1 finite elements.

Table 2: Consecutive manufactured-solution refinements. N_T is the triangle count and r_F is the free-system algebraic residual. All five levels use a sparse direct solve; an iterative stopping tolerance is not applicable at any level.

Level	N_T	DOFs	$h_{\max}$	$\|e\|_{L^2}$	$\|e\|_E$	$\|r_F\|_\infty$	p_{L^2}	p_E
2 (h)	32	25	0.35355	7.596×10^{-2}	8.397×10^{-1}	3.331×10^{-16}	–	–
3 ($h/2$)	128	81	0.17678	2.041×10^{-2}	4.320×10^{-1}	5.551×10^{-16}	1.896	0.959
4 ($h/4$)	512	289	0.08839	5.201×10^{-3}	2.176×10^{-1}	9.021×10^{-16}	1.972	0.989
5 ($h/8$)	2048	1089	0.04419	1.307×10^{-3}	1.090×10^{-1}	1.457×10^{-15}	1.993	0.997
6 ($h/16$)	8192	4225	0.02210	3.270×10^{-4}	5.451×10^{-2}	2.368×10^{-15}	1.998	0.999

Least-squares fits of $\log E$ against $\log h_{\max}$ over the finest three levels (4–6) give $p_{L^2} = 1.9956$ and $p_E = 0.9983$. These fitted orders complement the pairwise orders rather than selecting nonconsecutive meshes. The largest listed free residual is below 2.4×10^{-15}; the verification residual acceptance check is $\|r_F\|_\infty < 10^{-10}$, not an iterative solver tolerance.

4.2 Layered-interface analytic verification

The Problem Pack supplies a second check aimed specifically at discontinuous coefficients. On the dimensionless unit square, set $a = 1$ for $x < 1/2$ and $a = 10$ for $x > 1/2$, with $f = 0$ and exact Dirichlet data on the entire boundary:

$$u_*(x) = \begin{cases} 20x/11, & x \le 1/2, \\ (2x+9)/11, & x \ge 1/2. \end{cases}$$

Both traces equal $10/11$ at the interface, while the gradients are $20/11$ and $2/11$. Thus the physical normal flux is $q_* = -a\partial_x u_* = -20/11$ on both sides. This checks the coefficient-weighted flux, rather than equality of the raw derivatives.

The test calls the production `assemble_stiffness` routine with relative coefficients 1 and 10, then divides out the common ε_0 to obtain the dimensionless problem. Strong Dirichlet elimination is applied with the analytic boundary values. Because the fitted exact solution belongs to the continuous piecewise-affine space, errors should be at round-off; a convergence slope fitted to these tiny errors would have no useful meaning.

Table 3: Layered verification using the actual production assembly. Odd N deliberately leaves the interface unresolved and assigns coefficients by element centroid; it is a negative control, not an acceptable reference method.

N	Fitted?	DOFs	$\|u_h - u_*\|_\infty$ (nodes)	$\|q_h - q_*\|_{L^2}$
4	yes	25	3.33e-16	3.18e-15
8	yes	81	8.88e-16	1.04e-14
16	yes	289	2.11e-15	3.86e-14
32	yes	1089	3.22e-15	9.15e-14
7	no	64	8.15e-02	1.07e+00
15	no	256	3.95e-02	5.80e-01
31	no	1024	1.94e-02	3.42e-01

On fitted meshes the shared interface nodes enforce potential continuity, the maximum normal-flux jump is 2.28×10^{-13}, and the reduced matrices are symmetric with positive smallest eigenvalues. All solutions remain between zero and one. The unresolved control also has a small residual and a symmetric positive matrix, yet has much larger solution and flux errors. Algebraic checks alone therefore cannot establish correct interface physics.

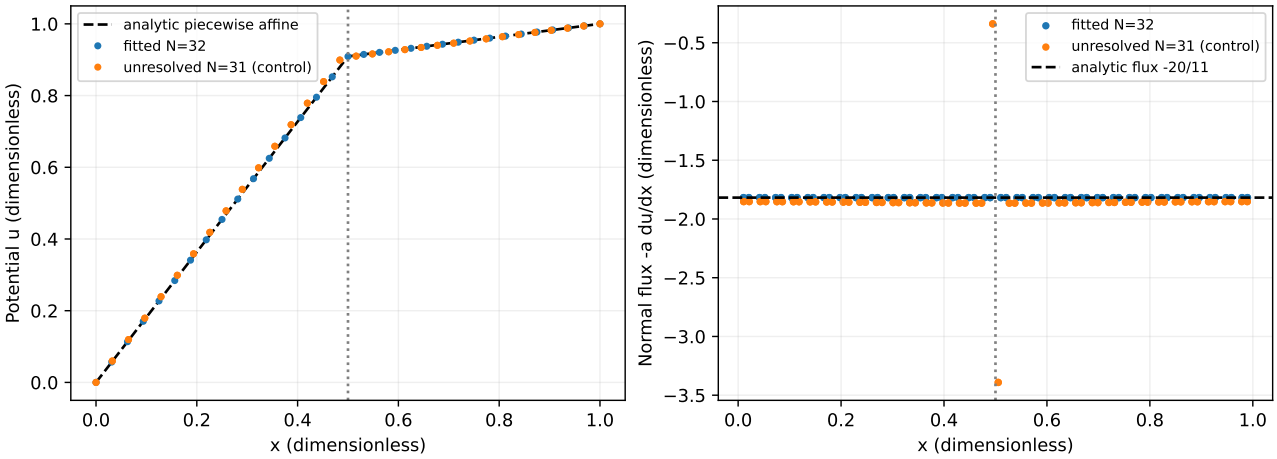


Figure 3: Analytic potential and normal flux across the two materials. The fitted solution reproduces both; the unresolved centroid-tag control shows why accurate interface representation must be checked separately.

4.3 Baseline microstrip sanity checks

The representative baseline solve passed the expected physical and algebraic checks. The potential remained in the interval $[0, 1]$, the symmetry error about $x = 0$ was near round-off, the reduced matrix symmetry error was zero to numerical precision, and the residual was negligible. The energy and reaction capacitances agreed to within 2.54×10^{-15} relative difference.

Linear solver and algebraic-error control. After Dirichlet elimination, the free system is sparse, symmetric and positive definite, so a sparse direct solve is appropriate for the tested problem sizes. This choice also avoids ambiguity about iterative stopping tolerances. A small residual alone would not prove negligible algebraic error, so the residual is interpreted together with conditioning diagnostics. For the representative baseline solve, the estimated physical reduced-system condition number was 1.7922363×10^{5}, while the diagonal-scaled diagnostic was 1.4249857×10^{4}. The free residual infinity norm was 3.101927×10^{-25}. The scaled diagnostic is not the physical condition number, but these conditioning and backward-residual diagnostics indicate that algebraic solve error is far below the observed finite-element discretisation effects.

Table 4: Representative baseline checks on the $8H$ microstrip domain.

Diagnostic	Value
Potential range	$[0, 1]$
Symmetry relative error	3.244×10^{-14}
Reduced-matrix symmetry error	0
Smallest eigenvalue estimate	5.918×10^{-15}
Largest eigenvalue estimate	1.060677×10^{-9}
Condition estimate	1.7922363×10^{5}
Diagonal-scaled condition diagnostic	1.4249857×10^{4}
Free residual infinity norm	3.101927×10^{-25}
C_E and C_Q diagnostic difference	2.536×10^{-15}
Charge-balance defect	6.973×10^{-15}

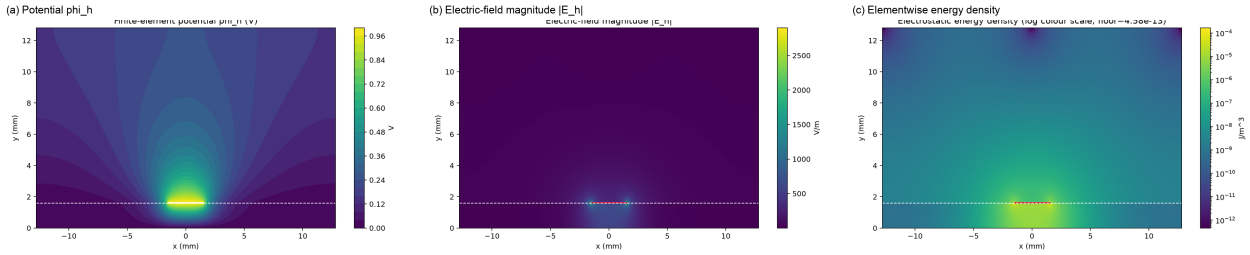


Figure 4: Baseline field diagnostics: (a) finite-element potential ϕ_h, (b) electric-field magnitude $|E_h|$, and (c) elementwise electrostatic energy density, shown with a logarithmic colour scale for visibility. The potential respects the imposed Dirichlet values, while the field and energy density concentrate near the trace corners. These plots are diagnostic visualisations, not error norms.

5 Domain Truncation

The physical air region is unbounded, but the computation uses a finite rectangle. To isolate the truncation effect, the domain study used the sequence $(W/H, H_{\text{air}}/H) = (3,3), (5,5), (8,8), (12,12)$ while preserving the near-trace mesh. The $12H$ result was used only as a tested finite-domain reference, not as the exact infinite-domain solution.

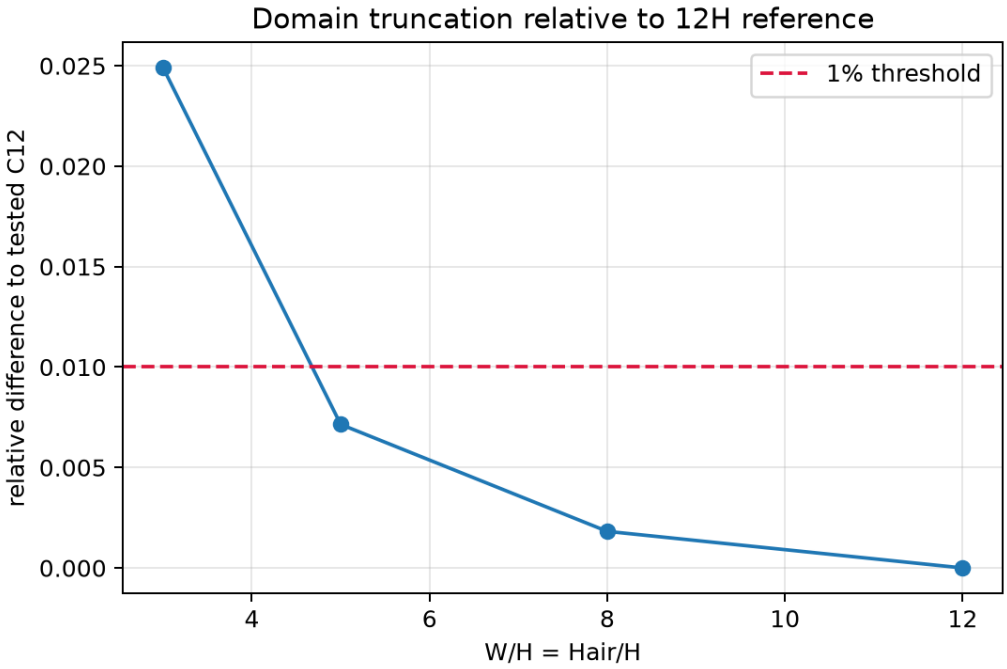


Figure 5: Domain-truncation study. Differences are relative to the tested $12H$ finite-domain reference. The horizontal line marks the one-percent threshold.

Table 5: Production domain-truncation results.

$W/H = H_{\text{air}}/H$	Elements	DOFs	C_E (pF/m)	Difference from $12H$
3	11868	6121	118.839	2.491%
5	16988	8711	121.004	0.714%
8	26168	13346	121.653	0.182%
12	41208	20926	121.875	reference

The largest-domain mesh-adequacy check repeated the $12H$ case with a finer near-field resolution.

The production value was $C_{12,R0} = 1.2187452011154754 \times 10^{-10}\,\mathrm{F/m}$, while the refined repeat gave $C_{12,R1} = 1.2183061087416544 \times 10^{-10}\,\mathrm{F/m}$. The resulting mesh sensitivity was 3.604122×10^{-4}, or about 0.0360%. This is smaller than the $8H$–to–$12H$ production difference of about 0.182%, so the selected-domain conclusion is not driven primarily by the production near-field mesh. Neither $12H$ value is treated as the exact infinite-domain capacitance.

The production $8H$ value in Table 5, 121.653 pF/m, and the final M7 value, 120.620 pF/m, use different mesh families and resolutions. Their difference cannot be attributed solely to truncation. The 0.182% indicator compares matched-core production-domain meshes; it is not a measured truncation error, or an upper bound, for M7. The finer $12H$ repeat strengthens the original domain decision but does not replace a matched-M7-resolution $8H/12H$ study, which has not been performed.

The smallest tested domain below one percent was $5H$. The later studies use the $8H$ domain because its tested difference from $12H$ was only about 0.182% while remaining substantially cheaper than the largest domain. This is a finite-domain sensitivity indicator rather than a rigorous infinite-domain error bound.

6 Mesh Convergence and Corner Singularity

6.1 Corner-field behaviour

Finite-element error estimates depend on solution regularity as well as the approximation space. Hackbusch [1, Secs. 8.5 and 9.1] and Hiptmair [2, Sec. 3.4] discuss this dependence and the effect of nonsmooth domains. These results explain why a smooth-problem rate need not apply here; they do not predict the measured value 0.623.

The conductor corners create singular electric-field behaviour. The lower trace corners are especially delicate because the conductor boundary meets the air–substrate interface. A mesh-dependent peak field that increases under refinement is therefore consistent with singular behaviour; it is not itself an accuracy norm.

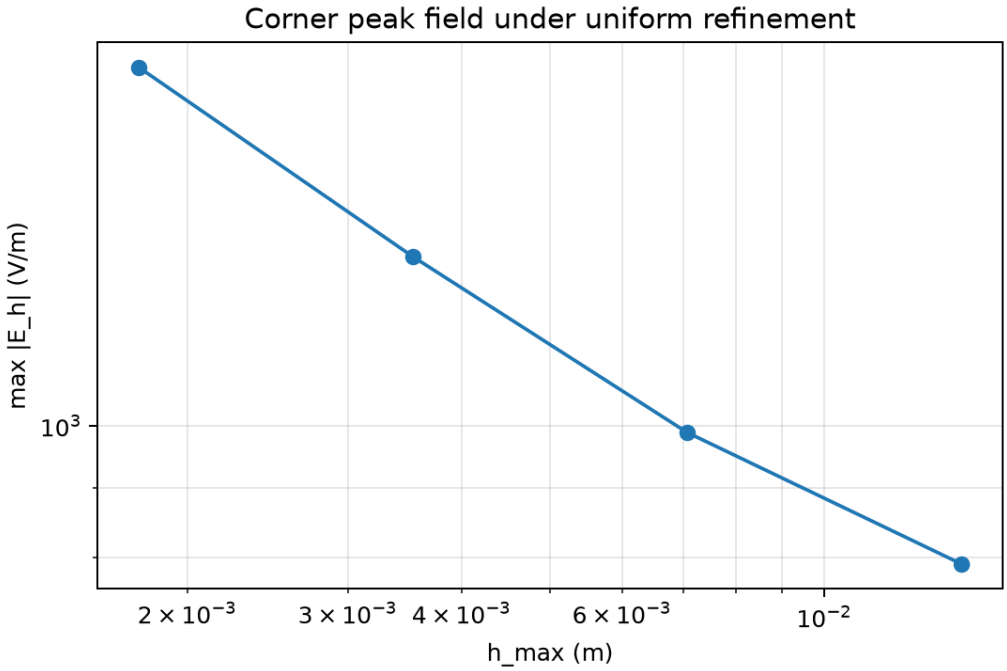


Figure 6: Maximum computed electric-field magnitude under uniform refinement. The increasing peak is consistent with stronger resolution of the trace-corner field.

6.2 Uniform energy-norm convergence

Uniform convergence was measured against an over-refined nested reference $R2 = M6$. The error norm was

$$\|e\|_E^2 = \int_\Omega \varepsilon |\nabla(\phi_{\text{ref}} - \phi_h)|^2 \, d\Omega.$$

The relative energy error used in the figures and tables is

$$E_{\text{rel}}^{\text{ref}} = \frac{\|\phi_{\text{ref}} - \phi_h\|_E}{\|\phi_{\text{ref}}\|_E}, \qquad \|\phi_{\text{ref}}\|_E^2 = \int_\Omega \varepsilon |\nabla \phi_{\text{ref}}|^2 \, d\Omega. \tag{3}$$

When the reference changes, both numerator and denominator use that same reference; this dimensionless ratio is not a relative capacitance difference. Coefficient vectors on different meshes were never directly subtracted. Instead, the coarse P_1 gradient was evaluated on the reference mesh elements and the error was integrated on the reference grid.

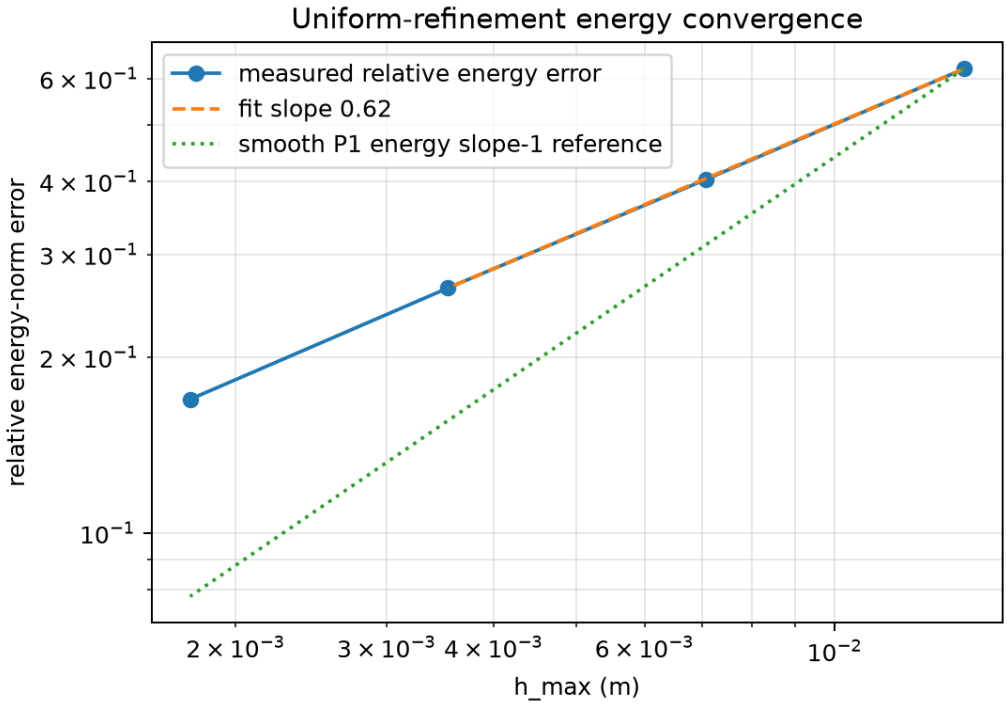


Figure 7: Uniform-refinement energy-norm convergence. The fitted slope over M0–M2 is about 0.623, below the smooth P_1 energy slope of one.

Table 6: Uniform energy-error convergence against the accepted $R2 = M6$ reference.

Mesh	DOFs	$h_{\max}$ (m)	Relative energy error	C_E (pF/m)
M0	35	1.415×10^{-2}	0.6230	167.703
M1	114	7.077×10^{-3}	0.4030	140.432
M2	404	3.538×10^{-3}	0.2627	129.144
M3	1512	1.769×10^{-3}	0.1694	124.273

Table 7: Reference stability in the M0–M2 fit window. R0=M4, R1=M5 and R2=M6. The last column is $|E_{\mathrm{rel}}^{R2} - E_{\mathrm{rel}}^{R1}|/|E_{\mathrm{rel}}^{R2}|$; each relative error is normalised as in (3).

Mesh	E_{rel}^{R0}	E_{rel}^{R1}	E_{rel}^{R2}	$R1 \to R2$ change
M0	0.610595	0.619193	0.623045	0.618%
M1	0.386757	0.398039	0.403041	1.241%
M2	0.239121	0.255580	0.262690	2.707%

The accepted fitted slope is 0.623 over M0–M2. The R1-to-R2 changes in Table 7 range from 0.618% to 2.707%; changing the reference changes the fitted slope from 0.6383 to 0.6230. Thus the qualitative sublinear-rate conclusion survives reference refinement. This finite sensitivity is disclosed rather than called zero or a rigorous negligible-error bound. M3 is excluded from the fit because its R1-to-R2 error sensitivity is larger (6.384%). The reduced rate is consistent with conductor-corner and material-interface regularity loss.

Fit-window robustness. Table 8 changes the reference and fit window without deleting any data. With R2, the adjacent M0–M1 and M1–M2 slopes are 0.6284 and 0.6176; adding M3 gives 0.6255, but M3 has the larger disclosed reference sensitivity. The three-point 0.623 fit is therefore a measured coarse-range trend, not a precise singular exponent or proof of an asymptotic limiting rate. R0 changes that same fit to 0.6762, showing why the over-refined reference matters.

Table 8: Existing-data sensitivity of the uniform energy fit. The accepted M0–M2/R2 fit is unchanged; the other rows test its interpretation.

Window	Reference	Points	Fitted h slope
M0-M2	R0	3	0.676239
M0-M2	R1	3	0.638307
M0-M2	R2	3	0.622987
M0-M3	R2	4	0.625473

7 Historical Graded Refinement and Integration Limitation

We tested whether concentrating coordinate-line refinement near the four trace corners could improve efficiency, with additional emphasis on the lower conductor/material-junction corners. The method remained symmetric about $x = 0$ and used the same physical model as the uniform sequence.

Reproducible grading rule. The shared base coordinate sets, in units of H, are

$$B_x/H = (-8, -2, -0.9375, 0, 0.9375, 2, 8), \qquad B_y/H = (0, 1, 1.021875, 1.5, 8).$$

Let $D_k(B)$ bisect every interval of B exactly k times (ratio 1/2). For parameters (b, u, ℓ), retain the background coordinates $D_b(B_x)$ and $D_b(B_y)$. Add coordinates from $D_\ell(B_x)$ within $0.45H$ of either $x = \pm w/2$ and from $D_\ell(B_y)$ within $0.45H$ of $y = H$. Add coordinates from $D_u(B_x)$ within $0.30H$ of either $x = \pm w/2$ and from $D_u(B_y)$ within $0.30H$ of $y = H + t$. Take sorted unions, snap to the M6 coordinate set at 10^{-15} m precision, form the Cartesian grid, and split each retained rectangle along its bottom-left/top-right diagonal; remove the trace interior. These are axis-aligned bands, not circular element-distance zones: adding a coordinate line also spends resolution away from a corner.

Table 9: Deterministic G-level parameters read from the experiment CSV. b is the background depth, u the upper-corner depth, and ℓ the lower-junction depth. The lower corners receive one extra depth for G0–G3; G4 uses depth five for both, with the larger lower-corner band retained.

Mesh	b	u	ℓ	DOFs	Min. corner size (m)
G0	0	1	2	123	4.210×10^{-4}
G1	1	2	3	537	2.105×10^{-4}
G2	1	3	4	1422	1.052×10^{-4}
G3	2	4	5	5306	5.262×10^{-5}
G4	3	5	5	7068	5.262×10^{-5}

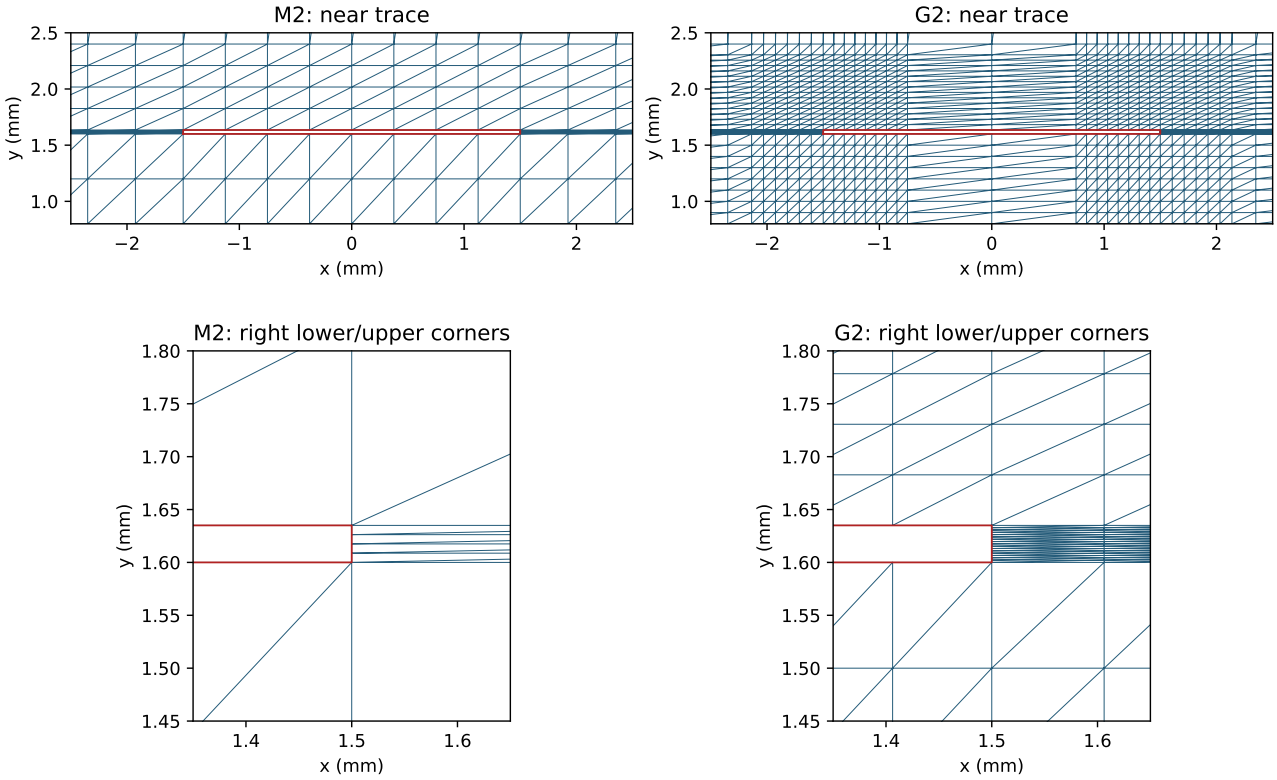


Figure 8: Saved uniform M2 and graded G2 meshes, with an additional right-corner enlargement. These are geometry views, not a matched-DOF efficiency claim; the three-method overlay comparison below uses the measured work metrics.

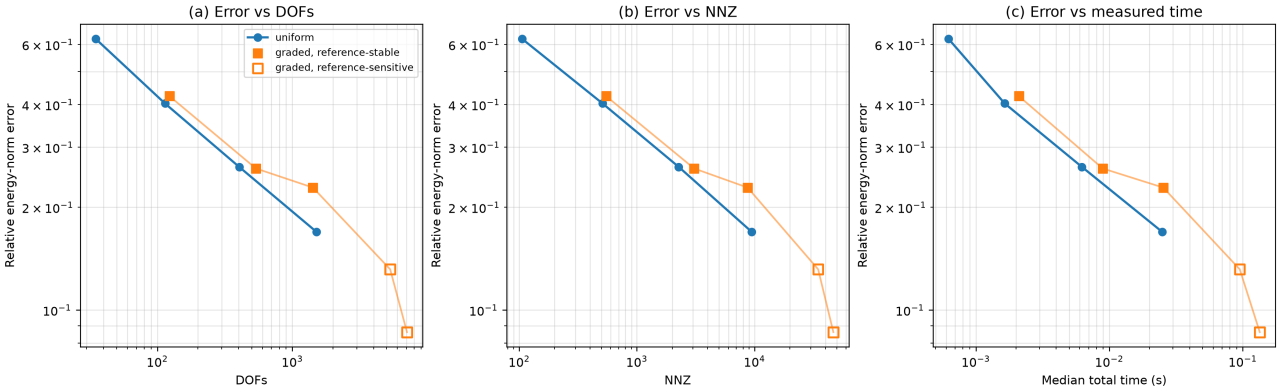


Figure 9: Historical exploratory estimates against (a) DOFs, (b) NNZ and (c) median time, retained unchanged. Filled orange squares pass the original reference-sensitivity check, but that check does not validate the cross-mesh integration. These curves do not support precise efficiency ratios; Figure 11 supplies the validated exact-overlay comparison.

We initially compared graded and uniform meshes using the saved energy-error estimates. The graded meshes resolved the corners more closely, but the estimates suggested no clear cost–accuracy advantage, including at targets 0.40 and 0.30. A later check showed that the cross-mesh integration was not always valid, so these initial comparisons remain exploratory.

Why the comparison was repeated. The coordinate-subset assumption did not ensure nested triangles. For G2, 21440 of the 180224 reference triangles cross a coarse diagonal, covering about 4.55% of field-domain area. This is not an error bound. We retained the original results, kept the rejecting safety guard, and repeated the comparison using triangle-overlay integration in Section 8. Appendix A, case audit 3 records the geometric test, replacement method and exact evidence locations.

As discussed by Hiptmair [2, Sec. 3.5], numerical quadrature can introduce additional errors into a finite-element calculation. In our original graded-mesh comparison, some reference triangles crossed coarse-element boundaries where the gradient was discontinuous. We therefore replaced the centroid-based error integration with triangle-overlay integration before making quantitative efficiency comparisons. Here the approximation affected error-norm post-processing, rather than the production stiffness assembly. The crossed-triangle count, 4.55% area fraction and corrected results are our own measurements, not conclusions taken from the text.

The initial lack of improvement also led us to test another refinement approach: choosing elements from the computed flux-jump residual rather than prescribing coordinate bands around the corners.

8 Residual-Based Adaptive Refinement

8.1 Motivation and controlled setup

The measured uniform energy rate, $p_E \simeq 0.623$, suggests that extra resolution near the strong corner fields may be useful. Rather than prescribing where to refine, we used the computed flux imbalance to select elements. We compared all three methods with the triangle-overlay calculation introduced above. The hypothesis was recorded before the experiments in `logs/PRE_ADAPTIVE_HYPOTHESIS.md`. The $8H$ domain, trace hole, material coefficient, homogeneous-Neumann far boundary, P_1 formulation and sparse direct solver were held fixed. No new all-air problem was solved, and the final M7 engineering values were retained.

The historical M2 coordinates are left–right symmetric, but its identically oriented cell diagonals are not. To avoid confusing mesh-orientation asymmetry with adaptivity, the primary initial mesh uses a conforming, mirror-invariant subdivision of M1. All edge midpoints are inserted; each right triangle is divided into four nonobtuse triangles, giving exactly the M2 coordinates, 704 elements and 404 DOFs. We call this *M1-S (M2-resolution)*, not the unchanged M2 triangulation. An eight-cycle

M0-S companion path brackets the 0.40 and 0.30 error targets, since the primary initial error is already below 0.30.

An earlier diagonal-only symmetry completion was rejected: its very coarse M0 companion produced minimum potentials down to about -0.030 V. All its points and timings remain in a separate failed-attempt directory and are excluded from the conclusions. The correction was made to satisfy potential bounds, not to optimise efficiency; the indicator and marking parameter were unchanged.

8.2 Flux-jump indicator and marking

A posteriori indicators use the computed solution to guide local refinement. The general background is given by Hackbusch [1, Sec. 8.7] and Quarteroni [3, Sec. 4.6]. These texts motivate the approach; the particular indicator, marking parameter and refinement closure tested here are stated below. We do not infer a guaranteed error bound or an optimal convergence rate from those references.

For $f = 0$, constant elementwise ϵ and affine $\phi_h|_K$, the interior volume residual vanishes. For an interior edge, choose one unit normal n_e and define

$$J_e = (\epsilon \nabla \phi_h|_{K^+} - \epsilon \nabla \phi_h|_{K^-}) \cdot n_e.$$

The same normal is used on both sides, including material-interface edges. With h_e equal to edge length, the implemented indicator is

$$\eta_K^2 = \frac{1}{2} \sum_{e \subset \partial K \cap \mathcal{E}_{\text{int}}} h_e \|J_e\|^2_{L^2(e)} + \sum_{e \subset \partial K \cap \Gamma_{\text{far}}} h_e \|\epsilon \nabla \phi_h \cdot n\|^2_{L^2(e)}, \qquad \eta = \left(\sum_K \eta_K^2 \right)^{1/2}. \tag{4}$$

Since these traces are constant, each term is evaluated exactly as $h_e^2 J_e^2$ (or its boundary counterpart). Trace and ground Dirichlet edges contribute no natural-boundary residual. This is a *residual-based refinement indicator*, not a guaranteed error bound.

At every step, deterministic Dörfler marking chooses the smallest descending-indicator prefix containing at least $\theta = 0.5$ of $\sum_K \eta_K^2$. The reflected triangles are added for symmetry. All edges of marked triangles are split; any triangle with two split edges has its third edge split as well. This conformity closure gives unambiguous red/four-child and green/two-child subdivisions. Minimal marks, symmetry-added marks and all actually refined parents are recorded separately. No corner-distance rule selects elements.

Algorithm 2 Residual-based adaptive refinement

Require: Symmetry-completed initial mesh; $\theta = 0.5$; eight cycles; M6/M7 references

1: **for** cycle $k = 0, \ldots, 7$ **do**
2: Solve the fixed dielectric problem using Algorithm 1.
3: Form $\varepsilon_K \nabla \phi_h$ and consistent-normal interior flux jumps.
4: Accumulate η_K^2, including far-Neumann edges and excluding Dirichlet edges.
5: Integrate errors on M6/M7 triangle overlaps; compute S_{ref} and C_E.
6: Record diagnostics, DOFs, NNZ and median timings after one warm-up and five runs.
7: Sort η_K^2 in descending order, breaking ties by element index.
8: Select the smallest prefix $\mathcal{M}$ with $\sum_{K \in \mathcal{M}} \eta_K^2 \geq \theta \sum_K \eta_K^2$.
9: Add reflected marks and close split edges for a conforming red/green subdivision.
10: Record minimal marks, symmetry additions and actually refined parents.
11: Verify hole, interface tags and conformity; use the refined mesh for the next cycle.
12: **end for**
13: Compare methods at bracketed targets; permit formal claims only if $S_{\text{ref}} < 0.03$ at every bracket endpoint.

Algorithm 2 uses the indicator to choose elements and the independently integrated reference error to judge accuracy. Outgoing refinement is charged to the next cycle's cumulative timing; offline

reference solves and overlay-error evaluation are excluded from every method's work metric. The primary M1-S and companion M0-S trajectories are run separately, not combined into a favourable envelope. The implementation is in `code/residual_indicator.py`, `code/adaptive_mesh.py` and `code/residual_adaptive_study.py`.

8.3 Reference error and adaptive mesh behaviour

The principal and sensitivity references are M6 and the frozen M7 dielectric solution. For $R \in \{M6, M7\}$,

$$\|e_R\|_E = \left(\int_\Omega \epsilon |\nabla(\phi_R - \phi_h)|^2 \, d\Omega \right)^{1/2}, \quad E_{\mathrm{rel},R} = \frac{\|e_R\|_E}{\|\phi_R\|_E}, \quad S_{\mathrm{ref}} = \frac{|\|e_{M7}\|_E - \|e_{M6}\|_E|}{\|e_{M7}\|_E}.$$

All three methods are compared using $E_{\mathrm{rel},M6}$ and the same exact triangle-overlay integration. Contained reference triangles use constant gradients directly; crossed triangles are intersected geometrically and integrated by overlap area. Here "exact" refers to integration of constant gradients over the geometric overlaps, subject to floating-point geometry tolerances, not to an exact PDE solution. Coefficient vectors on different meshes are never subtracted. This resolves the integration issue in Section 7 for the new comparison, while retaining its historical Figure 9 and raw tables. The maximum relative overlay area-coverage defect was below 2×10^{-12}. Formal comparisons require $S_{\mathrm{ref}} < 3\%$; other points remain visible with open markers.

Table 10: Primary residual-adaptive sequence. Every point meets the predeclared reference-sensitivity criterion. Full-precision capacitances, marks, sanity checks and timing samples are retained in the extension CSVs.

Cycle	Elements	DOFs	NNZ	E_{rel}	S_{ref} (%)
A0	704	404	2265	0.2637	0.847
A1	758	432	2439	0.2279	1.123
A2	898	504	2891	0.2056	1.325
A3	1218	666	3973	0.1789	1.610
A4	1650	890	5381	0.1670	1.600
A5	2270	1206	7469	0.1553	0.941
A6	3040	1602	10019	0.1434	0.297
A7	4008	2088	13345	0.1354	0.412

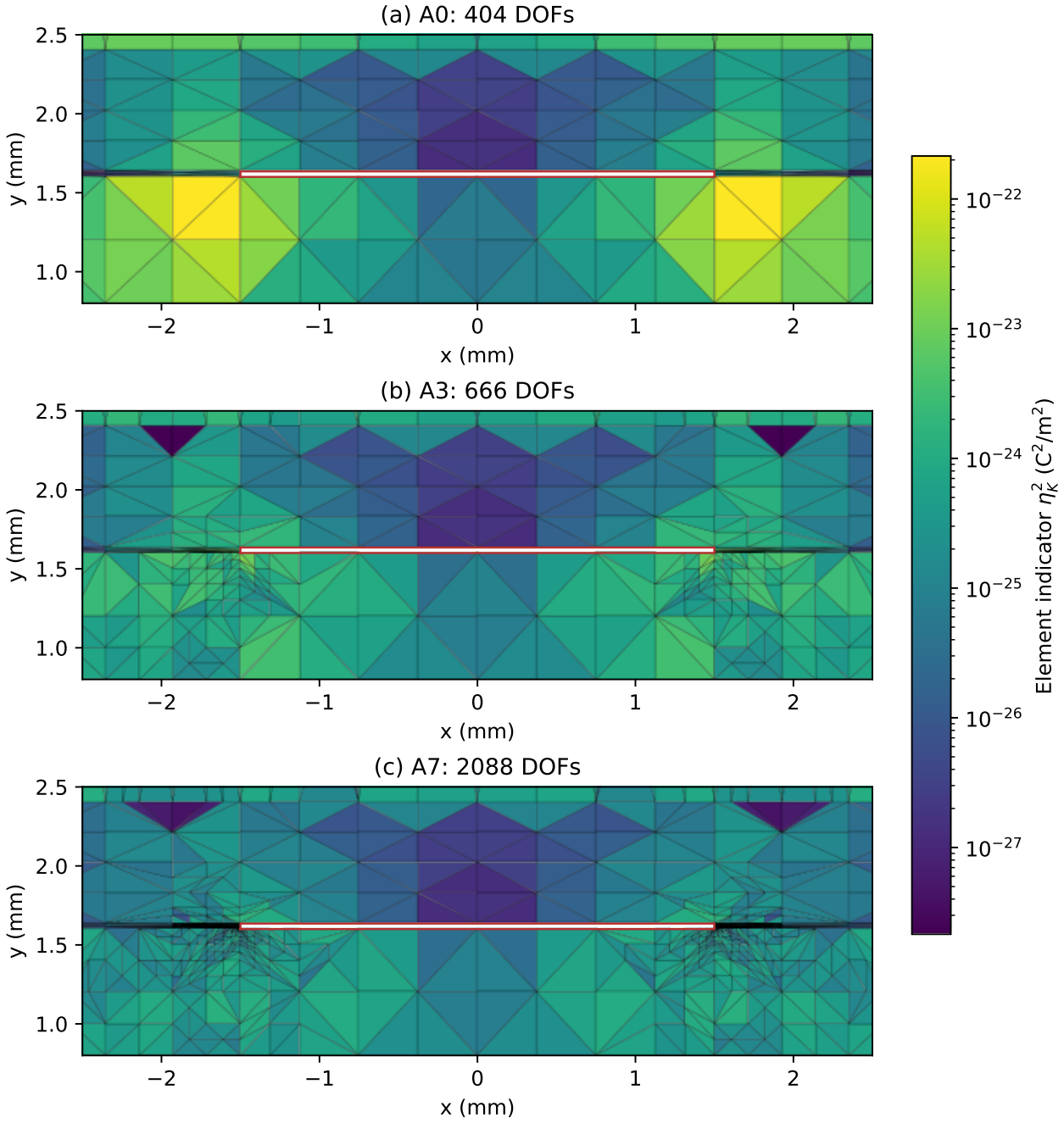


Figure 10: Residual-adaptive meshes A0, A3 and A7 in the same physical near-trace window; colours show η_K^2 on a shared logarithmic scale. Refinement increasingly concentrates around the trace ends, while some marked regions lie farther into the substrate and air. This is observed localisation, not a claim that the indicator exactly identifies every singular contribution.

Within $0.25H$ of a trace corner, the element fraction rises from 5.11% at A0 to 62.0% at A7; these elements contain 79.6% of the final squared indicator. Initially 88.6% of the indicator is assigned to elements whose nearest corner is a lower junction; at A7 this fraction is 50.2%. Thus lower-junction concentration is strongest early, not uniformly dominant throughout. The far-boundary contribution is about 0.00265% of the final indicator. Some final marked element centroids reach $|x| = 5.0$ mm and $y = 4.57$ mm, so refinement is not confined to four prescribed disks.

8.4 Matched-work comparison

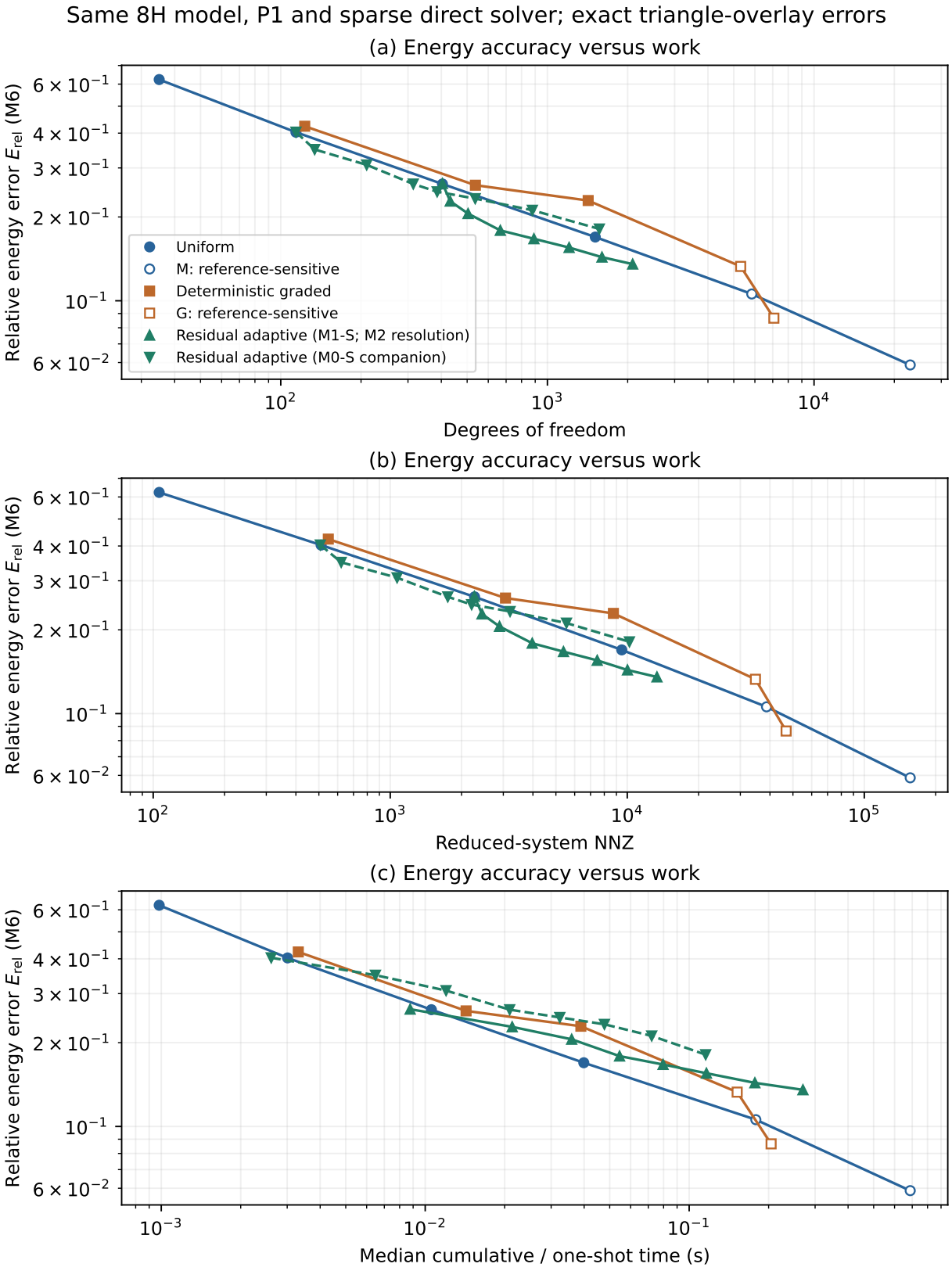


Figure 11: New three-way comparison in a common norm: (a) DOFs, (b) reduced NNZ, (c) measured median total algorithm time. Solid green triangles are the primary M1-S path; dashed green triangles are its separately identified M0-S companion. Open markers fail the M6/M7 reference-sensitivity threshold. Historical uniform/graded raw errors and timing tables were not overwritten.

One warm-up and five measured repetitions are used for every point. Uniform and graded work includes mesh generation, assembly, solve and capacitance extraction. Adaptive work is cumulative from its initial mesh, including all preceding solves, estimators, marking and refinement, up to the current solution. This accounting deliberately charges adaptive refinement for the preceding solve–estimate–mark–refine cycles because those computations are required to obtain the final adaptive mesh; the comparison is therefore conservative with respect to adaptivity. DOF and NNZ comparisons remain independent of this timing convention. Offline reference solves and error integration are excluded equally. This prevents comparing an entire adaptive trajectory with only a final sparse solve. Millisecond-scale differences on the coarsest meshes are not interpreted as convincing speedups.

Table 11: Piecewise log-linear interpolation between measured consecutive points only. U, G, A denote uniform, prescribed grading and residual adaptivity. Ratios above one favour adaptivity. The 0.40/0.30 adaptive targets use the M0-S companion. *The 0.20 three-way comparison is exploratory because the graded bracket includes reference-sensitive G3.

Target	N_U	N_G	N_A	N_U/N_A	$\mathrm{NNZ}_U/\mathrm{NNZ}_A$	T_U/T_A
0.40	116.6	146.2	115.1	1.013	1.022	1.115
0.30	272.8	348.7	224.0	1.218	1.235	0.546
0.20*	917.3	1974.2	532.7	1.722	1.791	0.618

At $E_{\mathrm{rel}} = 0.30$, adaptivity needs about 224 DOFs versus 273 for uniform and 349 for grading, with uniform/adaptive DOF and NNZ ratios 1.218 and 1.235. However, its cumulative median time is 13.1 ms versus 7.15 ms for uniform: the time ratio is 0.546, not a speedup. At 0.40, the DOF ratio is only 1.013 and the measured times are too small to establish a robust runtime advantage. In a direct reference-stable comparison, A5 uses 1206 DOFs with error 0.1553, while M3 uses 1512 DOFs with error 0.1694; A5 nevertheless takes about 2.91 times the cumulative algorithm time.

Measured interpolation brackets. The DOFs in Table 11 are interpolated work estimates, not newly solved meshes. Table 12 lists the actual consecutive meshes used in every interpolation. Targets 0.40 and 0.30 use the M0-S companion (B levels), whereas 0.20 uses the M1-S primary path (A levels) and remains exploratory because the graded bracket is reference-sensitive. The same log-linear weight is applied to each work metric; no extrapolation is used.

Table 12: Actual measured endpoints behind the target estimates. Sensitivity is the maximum over the two endpoints; the full CSV also records NNZ and interpolation weights.

Target	Method	Measured pair	DOFs (pair)	E_{rel} (pair)	max S_{ref} (%)
0.4	uniform	M1 / M2	114 / 404	0.4030 / 0.2627	1.107
0.4	graded	G0 / G1	123 / 537	0.4235 / 0.2601	1.138
0.4	adaptive B	B0 / B1	114 / 134	0.4036 / 0.3495	0.522
0.3	uniform	M1 / M2	114 / 404	0.4030 / 0.2627	1.107
0.3	graded	G0 / G1	123 / 537	0.4235 / 0.2601	1.138
0.3	adaptive B	B2 / B3	210 / 314	0.3077 / 0.2627	0.855
0.2	uniform	M2 / M3	404 / 1512	0.2627 / 0.1694	2.602
0.2	graded	G2 / G3	1422 / 5306	0.2291 / 0.1328	4.292
0.2	adaptive A	A2 / A3	504 / 666	0.2056 / 0.1789	1.610

Table 13: Median and interquartile range of selected original timing samples. Every level uses one warm-up and five measured repetitions. Adaptive totals are cumulative; fixed-mesh totals are one-shot. IQR describes repeat variability, not a confidence interval.

Mesh	Repetitions	Median total (ms)	IQR (ms)
A5	5	116.239	0.892
A7	5	269.791	18.892
B1	5	6.478	0.041
B2	5	11.995	0.027
M1	5	3.010	0.188
M2	5	10.554	0.513
G0	5	3.305	0.029
G1	5	14.285	0.032

The original sample files do not establish a complete hardware record. Their timings remain tied to that recorded run; the integrated reproduction records its own platform and memory preflight and produces new samples in a separate directory. Repeated measurements reduce single-run noise, but do not make millisecond-scale differences portable across machines. Nor does cumulative adaptive time isolate the speed of one sparse solve.

8.5 Interpretation and capacitance

We expected adaptive refinement to reduce the number of unknowns by refining the regions that needed more resolution. At 0.30, the results supported this expectation in terms of DOFs and matrix size. However, the repeated solving, indicator evaluation and refinement steps increased the total computation time. A7 has 1522 elements outside the specified corner neighbourhoods, versus 2047 for G2 at a comparable scale. This supports the explanation that coordinate-line grading spends more resolution away from corners; it does not isolate the causal effects of band width, lower-corner weighting or transitions, which were not varied independently.

The capacitance diagnostic uses $|C_h - C_{M7}|/|C_{M7}|$, where computed M7 is a selected numerical reference, not exact capacitance. A5 differs by 2.42%, versus 3.03% for M3, so the energy improvement also has a capacitance benefit in this matched-work pair. A7 still differs by 1.74% and is not substituted for the accepted M7 engineering mesh.

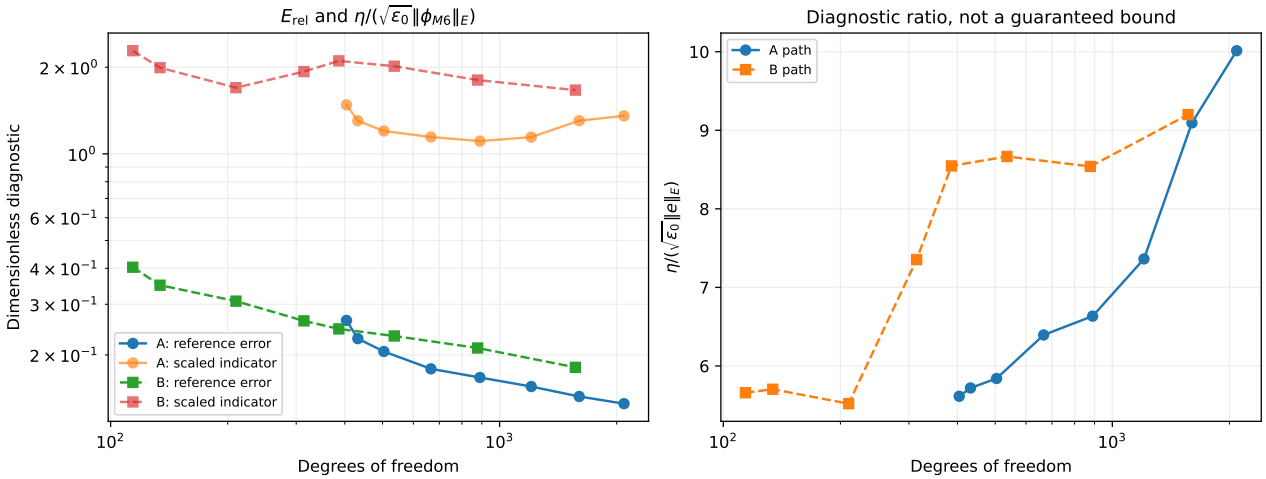


Figure 12: Indicator diagnostics. Physical η has units C/m, whereas $\|e\|_E$ has units $\sqrt{\text{J/m}}$. For a dimensionless display only, η is divided by $\sqrt{\epsilon_0}$ and the reference norm; the requested unscaled ratio $\eta/\|e\|_E$ is also stored. The dimensionless diagnostic ratio increases from about 5.61 to 10.0 on the primary path: it has not stabilised and is not an effectivity constant or guaranteed error bound.

The primary reference error decreases throughout, but η decreases initially and rises again over

the last levels. The indicator guides localisation in this experiment; the results are not sufficient to support a calibrated error-based stopping rule. All corrected A/B cycles satisfy potential bounds, symmetry, residual, matrix-symmetry, charge-balance, material and conformity checks; the failed first attempt remains explicitly archived.

What this comparison shows. Moving from fixed grading to residual-driven marking let the computed solution guide mesh refinement. We implemented the material-weighted flux-jump indicator, Dörfler marking and symmetry/conformity closure, and checked the errors using triangle overlays. At 0.30, the adaptive method used about 18% fewer DOFs and 19% fewer nonzeros than uniform refinement, but took about 1.83 times its cumulative algorithm time. The size reduction supports the localisation hypothesis; the timing result shows why it cannot be extended to a claim that adaptivity is always faster. This conclusion applies to the tested indicator, marking parameter and accuracy range.

8.6 Technical contributions and scope of novelty

Conforming P_1 FEM, residual flux-jump indicators and Dörfler marking are established methods; we do not claim to introduce them. The project-specific contribution is finding a faulty common-triangulation assumption in the graded comparison, replacing it with tested triangle-overlay integration, and using the same error definition and physical model to compare three refinement strategies. The negative timing result is part of this comparison: selecting a smaller final system can still cost more overall.

Does the experiment recover ideal DOF efficiency? For smooth two-dimensional P_1 approximation on shape-regular meshes, energy error of order h corresponds to $N^{-1/2}$. Fitting $\log E_{\mathrm{rel}}$ against $\log N$ gives Table 14. The primary M6-referenced slopes are -0.3620 over A0–A7 and -0.2470 over A3–A7. M7 gives the same qualitative conclusion. All points are retained, including the slower late range. These data do not demonstrate recovery of the ideal $N^{-1/2}$ rate; the observed DOF saving at one moderate target is a different statement.

Table 14: Empirical adaptive DOF slopes from existing data. These finite-window regressions are descriptive and do not establish estimator optimality.

Window	Reference	DOF slope	R^2
A0-A7	M6	-0.362027	0.9223
A0-A7	M7	-0.369620	0.9341
A3-A7	M6	-0.247016	0.9989
A3-A7	M7	-0.265684	0.9985

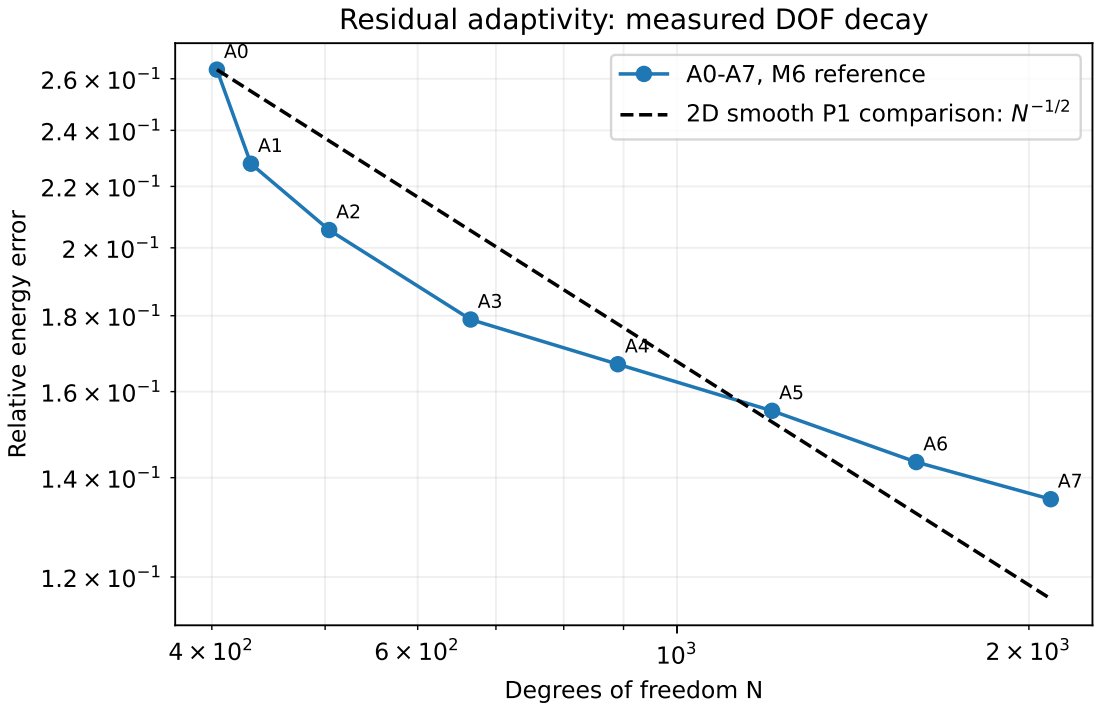


Figure 13: All primary adaptive points against DOFs. The $N^{-1/2}$ line is anchored at A0 only for slope comparison; it is not an error bound. The late decay is shallower.

Marking-parameter sensitivity. A separate eight-cycle study used $\theta = 0.3, 0.5, 0.7$, with the same M1-S start, model, M6/M7 references, five measured repetitions and cumulative accounting. All 24 cycles passed the geometry, potential, symmetry, residual, charge and overlay checks. The primary value remains 0.5; no earlier result was discarded or retuned.

Table 15: Supplementary marking study from the first integrated rerun. Equal cycle counts do not imply matched work. The full raw cycle data and time samples are retained separately.

θ	A7 DOFs	A7 E_{rel}	S_{ref} (%)	Cumulative time (ms)
0.3	1154	0.1605	0.446	145.3
0.5	2088	0.1354	0.412	237.7
0.7	4507	0.1055	1.750	409.0

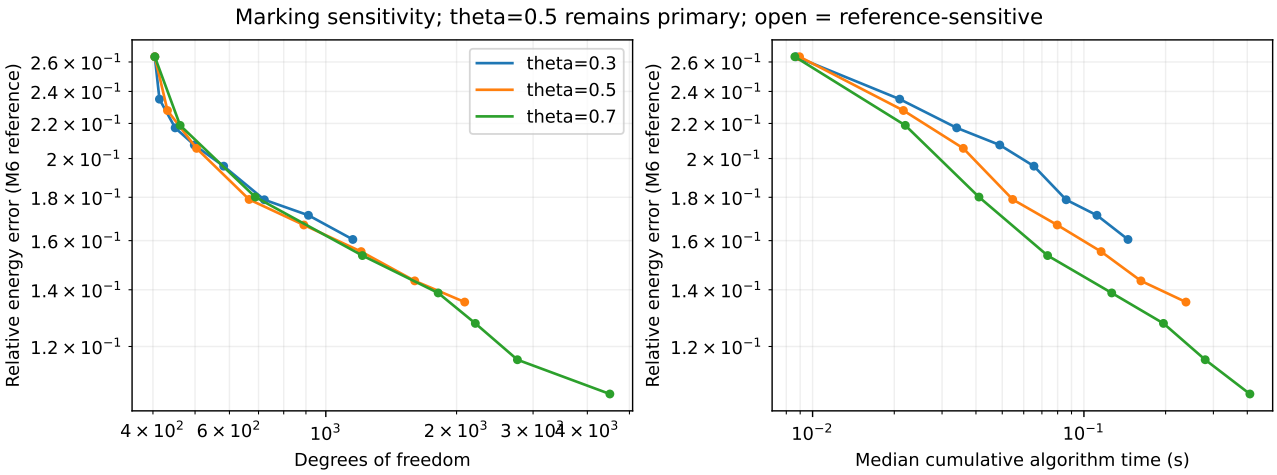


Figure 14: The same three marking parameters against measured DOFs and cumulative time. Larger θ refines more rapidly per cycle; that does not by itself establish better work efficiency. Open points are reference-sensitive.

9 Final Capacitance and Effective Permittivity

9.1 Final dielectric capacitance

The final capacitance selection used the nested uniform M3–M7 hierarchy on the fixed $8H$ domain. M8 was preflighted but not recommended because of the expected sparse-direct factorisation cost and fill-in. M7 was accepted through an asymptotic/resource fallback: the capacitance sequence decreased monotonically, the apparent output orders were stable, and the Richardson diagnostic was stable. The final reported capacitance remains the computed M7 FEM value, not the Richardson extrapolate. The M6–M7 change is 0.1556%, above the preferred 0.1% target. Acceptance therefore uses the stated project fallback criteria, rather than claiming that target was achieved. The following existing-data diagnostics show why the fallback is plausible, while leaving its finite spatial sensitivity visible.

Table 16: Output-order and Richardson diagnostics from all available successive triples. C_{ext} is never substituted for the computed M7 result.

Triple	p_C	C_{ext} (pF/m)	Finest-to-ext (%)	Limit change (%)
M3-M4-M5	1.17929	120.487011	0.6127	–
M4-M5-M6	1.15979	120.468826	0.2810	0.015095
M5-M6-M7	1.15524	120.466884	0.1269	0.001613

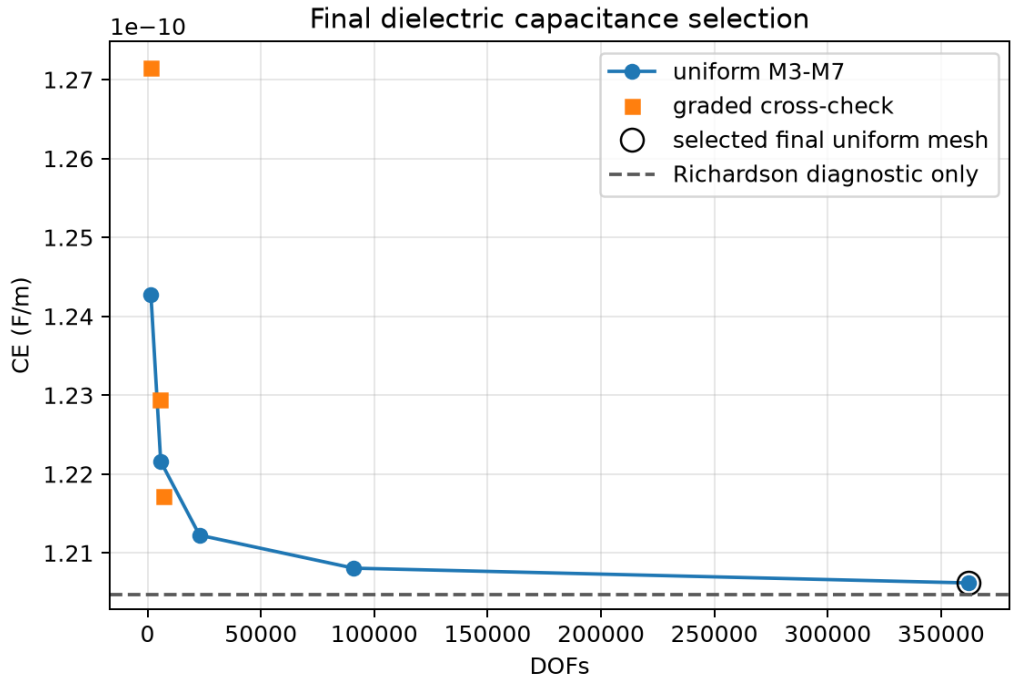


Figure 15: Final dielectric capacitance selection. M7 is the selected computed FEM mesh; the Richardson line is a diagnostic only.

Table 17: Final dielectric capacitance sequence.

Mesh	Elements	DOFs	C_E (pF/m)	Successive change
M3	2816	1512	124.273	–
M4	11264	5840	122.159	1.731%
M5	45056	22944	121.225	0.770%
M6	180224	90944	120.807	0.346%
M7	720896	362112	120.620	0.156%

The M7–M6 change, 0.1556%, is above the preferred direct 0.1% quality-control target. The reported value is the computed M7 result. Its remaining spatial sensitivity is approximately 0.156%, while the tested finite-domain sensitivity is approximately 0.182%; neither is a rigorous error bound. They are reported separately rather than combined statistically.

9.2 Matched all-air solve and effective permittivity

The all-air calculation reused the exact frozen M7 mesh. Only the material coefficient was changed to $\varepsilon_r = 1$ everywhere. This matching is important because $\varepsilon_{\text{eff}} = C/C_{\text{air}}$ is a ratio; changing the mesh would mix material and discretisation effects. The mesh fingerprint, degrees of freedom, boundary sets, and matrix sparsity pattern were all checked to match.

$$C = 120.620\,\text{pF/m}, \qquad C_{\text{air}} = 35.9682\,\text{pF/m}, \qquad \varepsilon_{\text{eff}} = \frac{C}{C_{\text{air}}} = 3.3535.$$

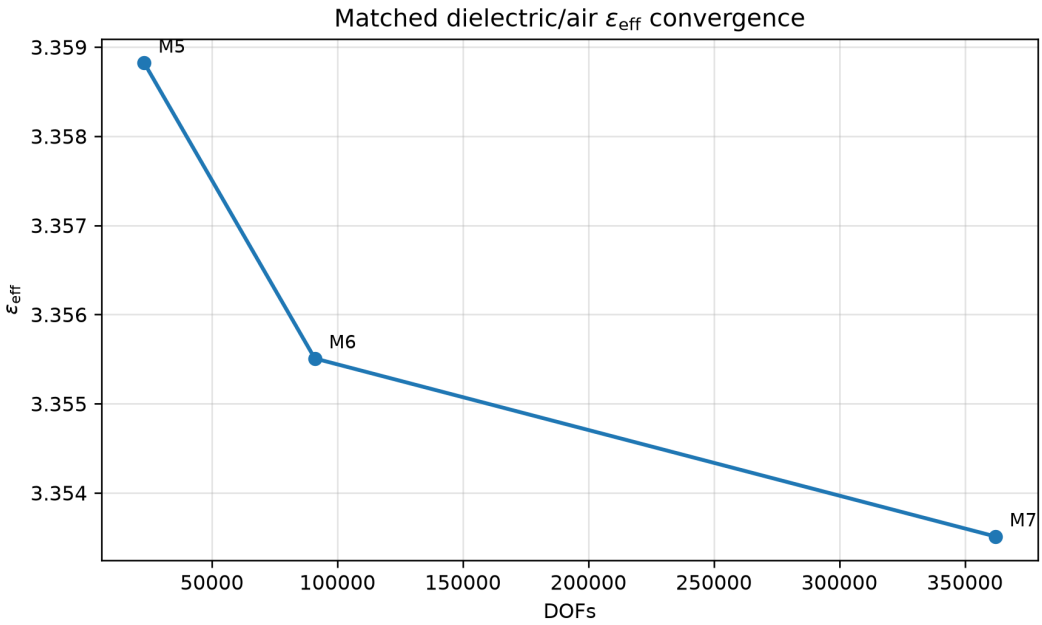


Figure 16: Matched M5–M7 effective-permittivity convergence. The M6–M7 relative change is about 0.0595%.

10 Characteristic Impedance and Engineering Validation

10.1 Final quasi-TEM impedance

Using the matched capacitance pair,

$$L = \frac{1}{c_0^2 C_{\text{air}}}, \qquad Z_0 = \sqrt{\frac{L}{C}} = \frac{1}{c_0 \sqrt{C C_{\text{air}}}},$$

with $c_0 = 299792458\,\text{m/s}$. The equivalent form

$$Z_0 = \frac{1}{c_0 C_{\text{air}} \sqrt{\varepsilon_{\text{eff}}}}$$

agrees to floating-point precision; this is an internal identity, not an independent validation.

Table 18: Final FEM transmission-line quantities.

Quantity	Value
C	$120.620\,\text{pF/m}$
C_{air}	$35.9682\,\text{pF/m}$
ε_{eff}	3.3535
L	$3.09343 \times 10^{-7}\,\text{H/m}$
v_p	$1.63708 \times 10^{8}\,\text{m/s}$
Z_0	$50.642\,\Omega$

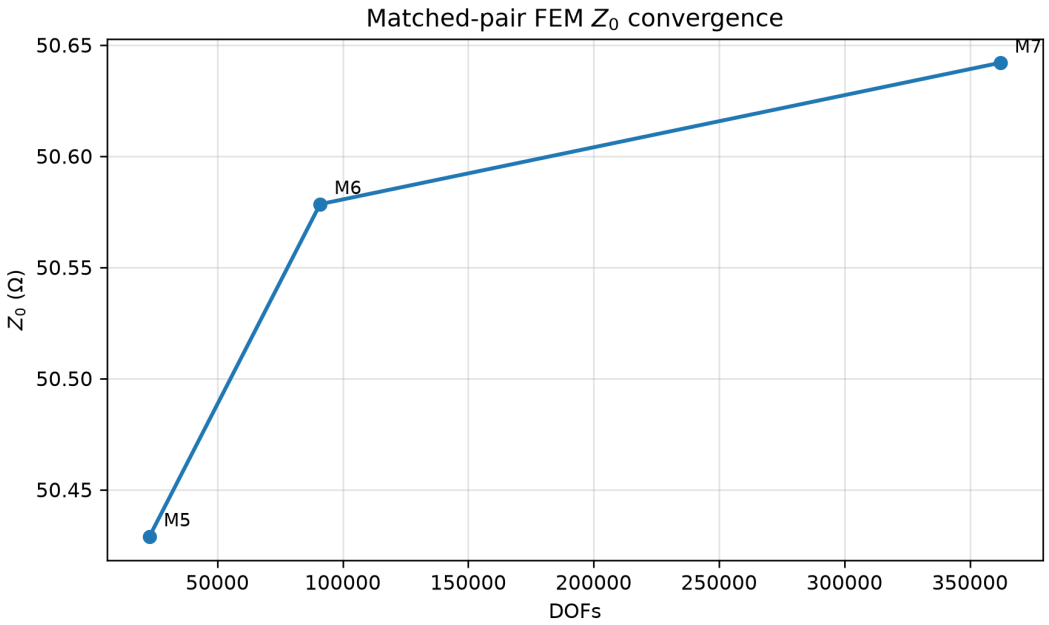


Figure 17: Direct matched-pair convergence of the final engineering quantity. The M6–M7 relative change in Z_0 is about 0.126%.

10.2 Independent engineering benchmarks

The final transmission-line table reports computed M7 outputs, with the 0.156% spatial and 0.182% finite-domain sensitivity caveats stated above; the printed digits do not establish an exact limiting value.

The Hammerstad–Jensen and IPC-2141 calculations use only geometry and ε_r, not FEM capacitance or FEM ε_{eff}. They are independent approximate engineering formulas, not exact solutions. The Hammerstad–Jensen implementation uses the finite strip-thickness correction from the project specification[2]; IPC-2141 uses the compact external-microstrip expression with a natural logarithm[3].

Following Appendix E of the specification, the named discrepancy uses Hammerstad–Jensen in the denominator:

$$d_{\text{HJ}}(Z) = \frac{|Z - Z_{0,\text{HJ}}|}{|Z_{0,\text{HJ}}|}.$$

The primary comparisons are therefore $100d_{\text{HJ}}(Z_{0,\text{FEM}}) = 0.879\%$ and $100d_{\text{HJ}}(Z_{0,\text{IPC}}) = 1.608\%$. The optional FEM-denominator differences are retained in a separately labelled column so that the two definitions cannot be confused.

[2] E. Hammerstad and O. Jensen, "Accurate Models for Microstrip Computer-Aided Design," *1980 IEEE MTT-S International Microwave Symposium Digest*, pp. 407–409, 1980, https://doi.org/10.1109/MWSYM.1980.1124303. The specific finite-thickness variant is the one documented in the supplied *Capacitor2d* appendix.

[3] IPC, *IPC-2141: Controlled Impedance Circuit Boards and High Speed Logic Design*, April 1996. We use the compact external-microstrip expression specified in the supplied *Capacitor2d* appendix, not a new transcription of the full standard. IPC confirms the original edition on the successor cover: https://www.ipc.org/TOC/IPC-2141A.pdf.

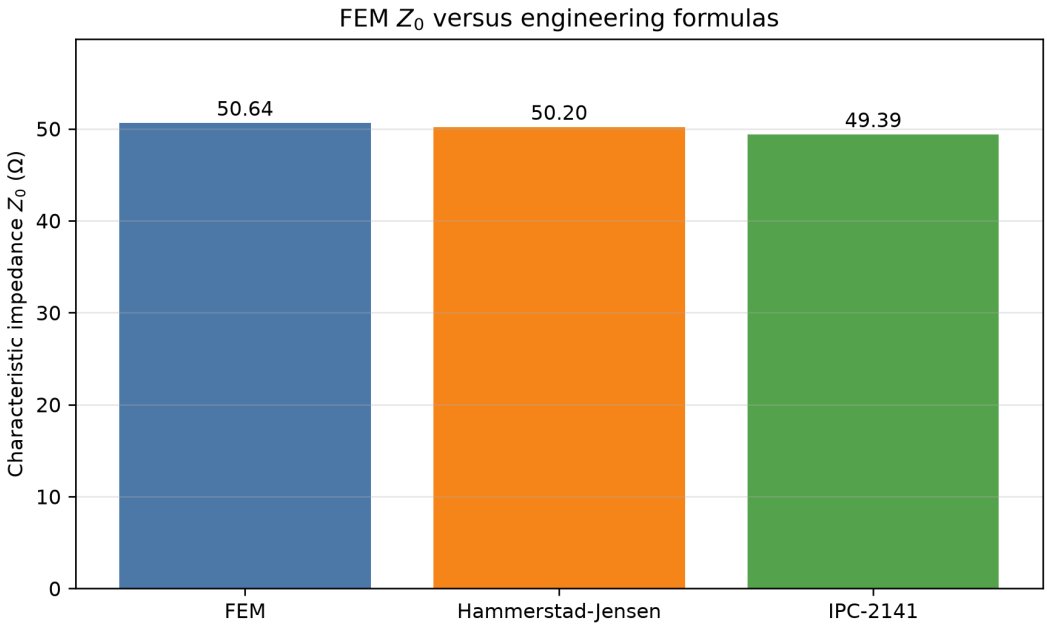


Figure 18: Final FEM characteristic impedance compared with independent engineering approximations.

Table 19: Engineering benchmark comparison. The primary d_{HJ} column uses the specification's Hammerstad–Jensen denominator. The last column is the supplementary $|Z - Z_{0,\mathrm{FEM}}|/|Z_{0,\mathrm{FEM}}|$ percentage, not d_{HJ}.

Method	$\varepsilon_{\mathrm{eff}}$	Z_0 (Ω)	$100d_{\mathrm{HJ}}$	Diff. vs FEM
FEM matched pair	3.3535	50.642	0.879%	0.000%
Hammerstad–Jensen	3.3008	50.201	0.000%	0.871%
IPC-2141	–	49.393	1.608%	2.466%

The Hammerstad–Jensen formula agrees more closely with the FEM result than the simpler IPC-2141 expression. This is useful engineering validation of the scale of the result, but the benchmark agreement does not prove FEM correctness. The numerical verification and convergence evidence above remain the primary validation.

The final FEM value is also close to the common 50 Ω design target:

$$Z_0 - 50\,\Omega = 0.642\,\Omega,$$

or about 1.284% above 50 Ω. The target is design context only, not a numerical reference solution.

11 Limitations

The physical model is electrostatic and quasistatic/quasi-TEM. It does not include dielectric dispersion, dielectric loss, conductor loss, surface roughness, solder-mask loading, radiation, anisotropy, or manufacturing tolerances. The trace and ground are perfect conductors.

The numerical result also has limitations. The $8H$ domain is justified by a finite $12H$ comparison, not by an exact infinite-domain solution. The final M7 mesh was accepted through a resource-aware asymptotic fallback rather than by directly reaching the preferred 0.1% capacitance-change target. Richardson extrapolation is diagnostic only. The historical graded curves use approximate cross-mesh quadrature; only the separately recomputed exact-overlay extension is used for current quantitative efficiency claims. The adaptive comparison uses one residual indicator, a primary Dörfler parameter ($\theta = 0.5$) with a separate 0.3/0.7 sensitivity study, one symmetry/conformity closure and two declared starting meshes. It does not establish optimality among estimators or marking strategies; the indicator-to-error ratio has not stabilised.

Reproduction commands and current scope

The pinned environment uses Python 3.13.13, NumPy 2.5.2, SciPy 1.18.1, Matplotlib 3.11.1, scikit-fem 12.0.2, Shapely 2.1.2 and pytest 9.1.1. The current machine is an Apple M2 with 16 GB RAM running macOS 26.6.2. Install the combined requirements and run from the repository root:

```
python3 -m pip install -r requirements-adaptive.txt
pytest -q
python3 code/run_all.py --quick
python3 code/run_all.py --full
```

Quick mode checks saved evidence and rebuilds plots in an isolated directory; it does not recompute the final M7 quantities. Full mode starts from numerical solves, replays the frozen mesh, and runs the overlay/adaptive comparison. It backs up the existing evidence and compares fresh output against it rather than overwriting it. The invalid graded shortcut still rejects crossed triangles; the integrated route explicitly selects polygon-overlay integration instead.

The first integrated run completed in about 190 s on this machine, passing 208 numerical comparisons with unchanged original protected files and the same final-mesh fingerprint. A later final-code run and its exact commands, resource preflight, tests and output manifests are recorded under `reproduction/` and `logs/ae_reproduction_report.md`. Regression testing currently reports 58 passed with no skips. This is evidence for this pinned local environment; a clean installation on a different operating system has not been executed. M8 is intentionally excluded. Timings vary: at the 0.30 target the first rerun gave adaptive/uniform cumulative time about 1.90, compared with 1.83 in the original experiment, preserving the qualitative runtime conclusion.

12 Conclusions

For the studied microstrip geometry, the selected M7 solution gives $C = 120.620\,\mathrm{pF/m}$. The all-air solve on exactly the same mesh gives $C_{\mathrm{air}} = 35.9682\,\mathrm{pF/m}$, leading to $\varepsilon_{\mathrm{eff}} = 3.3535$ and $Z_0 = 50.642\,\Omega$. This is close to the common $50\,\Omega$ design target.

At relative energy error 0.30, residual-based adaptivity needs approximately 224 DOFs, compared with 273 for uniform refinement and 349 for grading. It uses fewer DOFs and matrix nonzeros, but its cumulative time is 13.1 ms compared with 7.15 ms for uniform refinement. The preceding solve–estimate–mark–refine cycles add work that a one-shot uniform calculation does not require. Thus, fewer unknowns did not mean a shorter total runtime in this comparison.

The M6–M7 spatial indicator is approximately 0.156%, while the separate, coarser matched-core domain study gives 0.182%. Neither is a rigorous error bound, and the latter is not a measured M7 truncation error. M7 was accepted through the documented resource/asymptotic fallback, and Richardson extrapolation remains diagnostic. The quasi-static model also omits losses and full-wave dispersion or radiation, so the reported impedance should be interpreted within those assumptions.

References

[1] Wolfgang Hackbusch. *Elliptic Differential Equations: Theory and Numerical Treatment*, volume 18 of *Springer Series in Computational Mathematics*. Springer, 2 edition, 2017.

[2] Ralf Hiptmair. Numerical methods for partial differential equations. Lecture notes, ETH Zurich, 2025. Spring Term 2025, course 401-0674-00L.

[3] Alfio Quarteroni. *Numerical Models for Differential Problems*, volume 16 of *MS&A – Modeling, Simulation and Applications*. Springer, 3 edition, 2017.

A AI Transparency Audit

AI/Codex assistance supported code scaffolding, plotting, workflow generation, documentation organisation, consistency checks and report drafting and revision. Acceptance of numerical claims depended on manufactured-solution tests, matrix and residual diagnostics, domain/refinement studies, matched-mesh replay and independent formula checks. The three detailed audits below identify the assumptions, falsifiers, evidence locations and decisions for consequential claims; the final two audits summarise additional checks. A claimed audit outcome is distinguished from proof of individual authorship or complete pipeline certification.

The audit also follows the five-gate process required for AI-assisted solver work. First, formula and equation transcription was checked against the project source documents. Second, structure and indexing risks were tested through geometry, material-region, boundary-node, and dof checks. Third, the mathematical formulation was checked by deriving the weak form, Dirichlet elimination, capacitance identities, and quasi-TEM identities explicitly. Fourth, independent reproduction was addressed by the manufactured-solution test, mesh-refinement studies, matched air-only replay, and engineering benchmark formulas. Fifth, suspicious or fragile results were interrogated through the dedicated domain, reference-stability, graded-refinement, final-capacitance, and matched-mesh studies rather than accepted at face value.

Table 20: Project-specific AI audit summary.

Component	Main risk	Verification evidence
P1 FEM solver and Dirichlet elimination	Incorrect assembly, boundary treatment, or material coefficient handling	Manufactured-solution convergence; matrix symmetry and residual checks; material-tag tests
Microstrip geometry	Trace hole, interface, or boundary nodes misidentified	Geometry tests; full and near-trace figures; potential bounds and symmetry
Capacitance extraction	Reaction sign or trace-dof summation error	C_E/C_Q agreement and charge-balance diagnostics
Domain study	Changing the far domain could also change the near-trace mesh	Automated near-field equivalence checks and domain-truncation tables
Uniform convergence	Invalid coefficient-vector subtraction across meshes	Reference-mesh gradient transfer and reference-stability table
Graded refinement	Confusing coordinate nesting with triangular common refinement	Geometry audit detects crossed diagonals; original efficiency estimates retained with an explicit integration caveat
Final capacitance	Selecting a convenient mesh rather than an evidenced one	M3–M7 monotonic sequence, M8 preflight, Richardson diagnostic, frozen mesh artifact
Matched all-air solve	Regenerating a nominally similar mesh and contaminating C/C_{air}	Frozen mesh SHA-256, same dofs, same boundary sets, same sparse pattern, dielectric replay
Engineering benchmarks	Circular validation using FEM outputs inside formulas	Benchmark function signatures accept only geometry and ε_r; unit-invariance tests

Case audit 1: energy–charge agreement. *Claim examined:* near-equality of C_E and C_Q proves spatial convergence.
Assumptions: both quantities use the same discrete potential, stiffness operator, prescribed voltage and material-conforming element integrals.

Risk and falsifier: reactions and energy come from the same stiffness matrix, so even a coarse mesh can satisfy $2W_e = V_0 Q$.
Independent evidence: M0 has $\delta_{EQ} \approx 4.62 \times 10^{-16}$ but $C_E \approx 167.703\,\mathrm{pF/m}$, whereas the computed M7 result is $120.620\,\mathrm{pF/m}$.
Evidence locations: `tables/uniform_convergence.csv` (M0), `tables/final_capacitance_convergence.csv` (M3–M7), and `logs/gate_F_capacitance.txt`; the extraction is `extract_capacitance` in `code/capacitance.py`, with regression checks in `code/tests/test_final_capacitance.py`. The separate refinement sequence supplies the falsifying evidence, rather than another rearrangement of the same matrix identity.
Decision: reject that inference; use δ_{EQ} for algebraic consistency and the separate mesh sequence for spatial sensitivity.
Remaining limitation: the mesh sequence still has the disclosed M7 fallback sensitivity.

Case audit 2: a finite far-boundary reference. *Claim examined:* the tested $12H$ solution is the exact infinite-domain answer.
Assumptions: all four domains retain homogeneous Neumann outer boundaries and the same near-field mesh; only the outer region is extended.
Risk and falsifier: a finite Neumann boundary still changes the field; one large box alone cannot remove truncation.
Independent evidence: the fixed-core 3H–12H sequence and the finer 12H repeat separate measured domain and mesh effects.
Evidence locations: `tables/domain_truncation.csv`, `tables/domain_mesh_adequacy.csv` and `logs/gate_C_domain.txt`; generation is in `code/domain_study.py`, and fixed-core/domain checks are in `code/tests/test_domain_study.py`. Figure 5 displays the production comparison; Table 5 and its following paragraph report the fine-domain check.
Decision: call 12H a tested finite-domain reference and report the 0.182% 8H difference separately.
Remaining limitation: no exact infinite-domain error bound is available. This is an audited inference, not a claim that such an exact-reference error was previously committed.

Case audit 3: errors across different meshes. *Claim examined:* coefficient vectors or nested coordinate lists alone define a valid cross-mesh energy error.
Assumptions: the original shortcut treated every reference triangle as lying within one coarse triangle, so both gradients were constant on the integration cell.
Risk and falsifier: vector entries live in different bases, and a fine triangle crossing a coarse gradient jump invalidates constant-gradient element integration.
Independent evidence: uniform triangle-containment checks pass, while the new vertex-side test finds crossings in all five graded cases despite coordinate-subset tests passing.
Decision at the rubric review: retain the uniform norm calculation, correct the graded guard, and withdraw precise graded efficiency claims until overlay integration is validated.
Extension verification: Section 8 subsequently supplies separately stored exact-overlay errors for all three methods. Opposite-diagonal analytic integration, the nested Galerkin identity, material matching and area-coverage tests support that implementation. Historical curves were not overwritten.
Evidence locations: `tables/graded_reference_geometry_audit.csv` records crossings; `common_refinement_diagnostics` in `code/graded_study.py` checks them. The replacement integration is `overlay_energy_error` in `code/overlay_energy.py`. Its independent tests are `test_exact_overlay_crossing_diagonals` and `test_overlay_reproduces_uniform_nested_energy_identity` in `code/tests/test_residual_adaptive.py`. Current errors and formal targets are in `results/adaptive_extension/uniform_graded_overlay_comparison.csv` and `results/adaptive_extension/adaptive_matched_targets.csv`; `logs/adaptive_final_validation.txt` records the standalone run validation.
Remaining limitation: fine graded points remain reference-sensitive. The revised unified entry now uses overlays and has been rerun; the historical centroid shortcut remains blocked. Cross-platform reproduction remains untested.

Case audit 4: symmetry and mesh-quality assumptions. *Claim examined:* symmetric node coordinates alone ensure a symmetric continuous P_1 field after local refinement.
Falsifier and evidence: historical parallel diagonals gave an approximately 2.05% continuous reflection discrepancy on M2 despite near-roundoff nodal symmetry. A first diagonal-only completion then failed coarse-companion potential bounds.
Decision: retain the failed attempt, use a nonobtuse mirror completion of the same parent family, and test continuous reflection, bounds, conformity and material tags independently. Corrected paths pass; this is a validation-driven correction, not tuning θ to win.

Additional coefficient and reproduction checks. The layered analytic test uses the production assembly, while its unresolved-interface control shows that small residuals and an SPD matrix can coexist with a wrong material model. Evidence is in `code/layered_verification.py`, `code/tests/test_layered_verification.py` and `results/ae_revision/layered_interface_verification.csv`. The renewed reproduction is checked by `code/reproduction.py` and `code/tests/test_reproduction.py`; a failed command or changed frozen value is an error, not a successful run. Raw numerical comparisons and resource decisions are retained under `reproduction/`. No Git development history was manufactured; the available history remains dominated by upload commits.

Case audit 5: estimator interpretation. *Claim examined:* a residual indicator is automatically a calibrated true-error estimate.
Falsifier and evidence: the primary reference error decreases, while the final indicator rises and the scaled indicator/error ratio changes from about 5.61 to 10.0.
Decision: use η for marking only; report its dimensional and scaled diagnostics without a guaranteed-bound or stable-effectivity claim. Efficiency claims use independently integrated reference errors, not η.

B Individual Contribution Statements

Member 1:
Chuqiao Tang (2360692). **[TO BE COMPLETED BY TEAM.]**

Member 2:
Jinyu Ma (2363433). **[TO BE COMPLETED BY TEAM.]**

Member 3:
Muqing Xu (2362900).

My main responsibility in this project was code reproduction and numerical validation. I focused on understanding and reproducing the FEM code developed by the team. I reviewed the project structure, set up the Python environment, and followed the instructions in `README.md` to run the available programs. I also used the test scripts in `code/tests/` to check whether the implementation worked correctly.

During reproduction, I compared the computed results with the report. This covered domain-size studies, mesh convergence, capacitance and characteristic impedance. I checked whether the figures and numerical outputs were consistent with the original results.

When I encountered execution problems, I examined the code, dependencies and input settings to identify possible causes. I shared the issues and my findings with the team to help improve the reliability and reproducibility of the project.

My main contribution was independently checking the existing implementation and its numerical results. This helped the team identify potential problems and provided additional confidence in the accuracy and consistency of our work.

Member 4:
Shuoran Fu (2362626).

I worked on developing and organising the Python FEM code, including mesh generation, boundary conditions, sparse matrix assembly and solution of the electrostatic problem. My work covered files such as `code/geometry.py`, `code/fem_solver.py` and `code/run_all.py`. I also worked on numerical experiments involving domain size, mesh refinement, convergence analysis and adaptive refinement. I examined the results to identify numerical issues and improve the reliability of the implementation.

For the report, I designed the overall structure and contributed to writing and revising the technical sections in `report/main.tex`. These sections explain the mathematical model, FEM method, verification process, numerical experiments and conclusions. I organised figures and tables, checked numerical values, and revised the discussion to reflect the limitations identified during testing.

AI tools assisted with coding, debugging and writing. I reviewed the generated content and checked important results against the mathematical model and numerical evidence. My main contribution was connecting the implementation, numerical experiments and report into a consistent project.

Repository evidence: commits `2b9dfed` and `b5eee20` record the code and report uploads under Shuoran Fu's account; `7119103` records the frozen M7 mesh addition. These are delivery records supporting the supplied statement, not a complete record of individual development steps.

Member 5:
Taiqi Zhang (2363684). **[TO BE COMPLETED BY TEAM.]**

Member 6:
Yujia Ding (2362094).

My main responsibility in this project was the mathematical theory and formulation of the microstrip FEM problem.

I worked on converting the physical microstrip structure into a two-dimensional electrostatic boundary value problem. This included defining the governing PDE, material properties and boundary conditions. I also helped derive and check the weak formulation and the P1 finite element discretisation.

During the project, I focused on understanding how the dielectric interface is handled in the weak formulation and checking the mathematical expressions used for capacitance, effective permittivity and characteristic impedance. I also reviewed the assumptions behind the numerical model and the properties of the stiffness matrix.

For the numerical analysis, I helped interpret the convergence results, particularly why the energy convergence rate was lower than the theoretical rate for a smooth problem. I considered how conductor corners and material interfaces could affect the solution.

I also checked the mathematical equations and explanations in the report to make sure they were consistent with our numerical methods and results.

My main contribution was providing the mathematical foundation for the FEM implementation and helping the team explain the theoretical reasoning behind our numerical experiments.